Government of Nepal
National Reconstruction Authority
Singhadurbar, Kathmandu

HYBRID STRUCTURE MANUAL

for

houses that have been built under the
HOUSING RECONSTRUCTION PROGRAMME

2017

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PART-1: Background and Characteristics of Hybrid Structure

Background

On 25th April, an earthquake of magnitude 7.8 struck with epicenter in Barpak, Gorkha. Where several aftershocks were still being felt, meanwhile another major aftershock hit Dolakha district on 12th May, 2015.
The most damage and destroy house were constructed by low strength masonry building with mud mortar houses, especially two story building that predominate in the rural area in Nepal.

Under Housing reconstruction programme, in order to make the earthquake resistance buildings that are to be constructed, NRA has formulated a minimum requirements (MRs) based on the NBC 105.
The MRs clearly stated that for the buildings with stone masonry in mud mortar, the number of story is restricted to only one story if wooden band is used, whereas if RC band is used, allowable number of story is one story plus attic, based on structural analysis.

Nevertheless, the people tends to construct two story building to meet their living functional requirement. Hence, they have started to build a light weight structure such as timber or steel with CGI sheet wall and roof, above masonry building. And one other reason is they have experience of risk of masonry in higher position in the wall. Hybrid structure is huge demand in the reconstruction field.

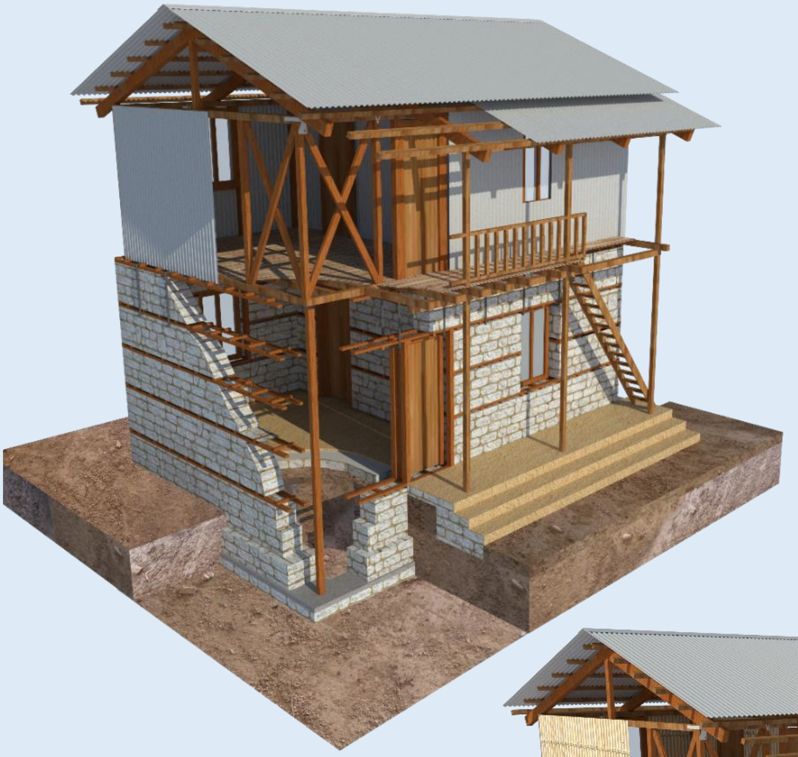
People have built the upper story mostly timber frame structure with the available knowledge and local materials.

Hence in order to ensure the safety of these building against wind load and earthquake load, it has become an urgent task to make the construction guideline along with proper connection details and standards of hybrid structures.

Definition of Hybrid structure

Hybrid structure (Mix structure is called in academic field) is the combination of two or more type of structural system that is generally constructed with different technology and materials in accordance with level of floor.

The hybrid structure in Nepal reconstruction program, the ground floor is constructed by masonry structure, and the first floor is usually constructed of lightweight materials such as CGI sheet with timber or steel frame structure.










SCOPE

❑ Applicability

This Guideline/Manual is prepared on the basis of NBC105, IS codes and NBC104 wind load.
The designs mentioned in the manual are ready-to-use designs for all structural components, but some provisions mentioned are set as advisory measures..

❑ Limitations

This manual covers only hybrid structure under the GoN housing reconstruction programme.
The ground floor is constructed by masonry structure, and the first floor is usually constructed of lightweight materials such as CGI sheet with timber or (steel) frame structure.

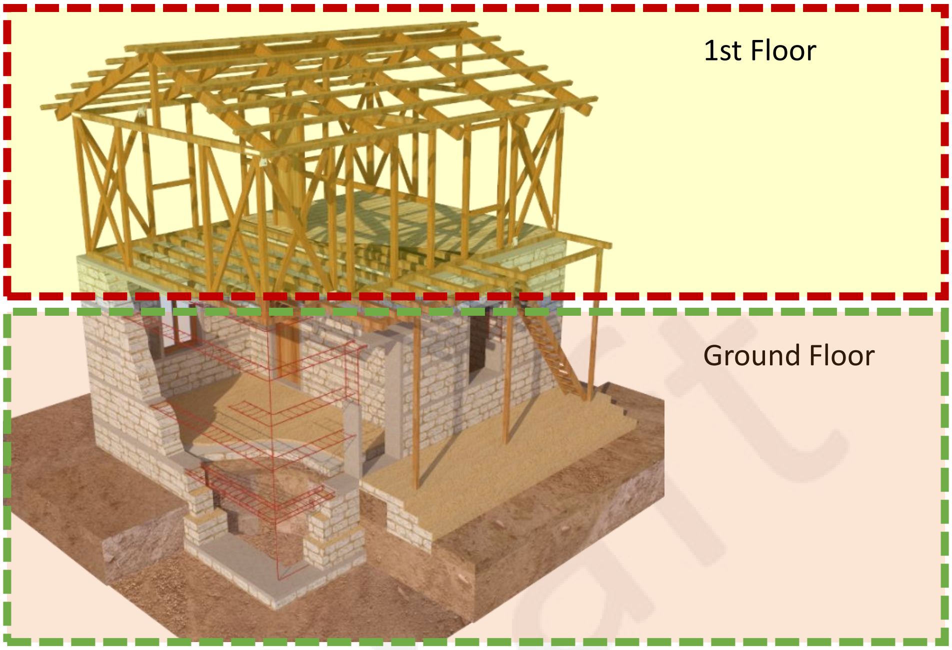
This manual has certain limitations and is only relevant for buildings which are:

I. Residential and fall under category 'C' and 'D' of NBC.

- ✓ Category "A": Modern building to be built, based on the international state-of-the-art, also in pursuance of the building codes to be followed in developed countries.
- ✓ Category "B": Buildings with plinth area of more than One Thousand square feet, with more than three floors including the ground floor or with structural span of more than 4.5 meters.
- ✓ Category "C": Buildings with plinth area of up to One Thousand square feet, with up to three floors including the ground floor or with structural span of up to 4.5 meters.
- ✓ Category "D": Small houses, sheds made of baked or unbaked brick, stone, clay, bamboo, grass etc., except those set forth in clauses (a), (b) and (c)

Limitation of this manual

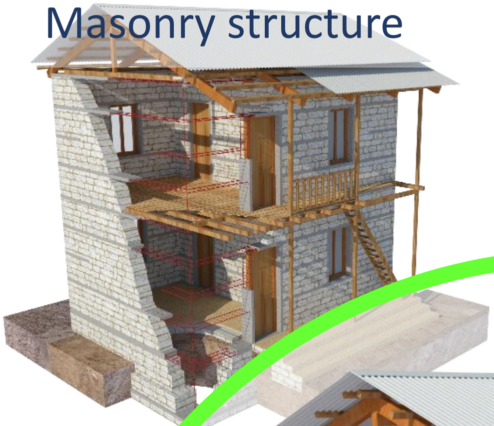
Hybrid structure (Mix structure is called in academic field) is the combination of two or more type of structural system that is generally constructed with different technology and materials in accordance with level of floor.



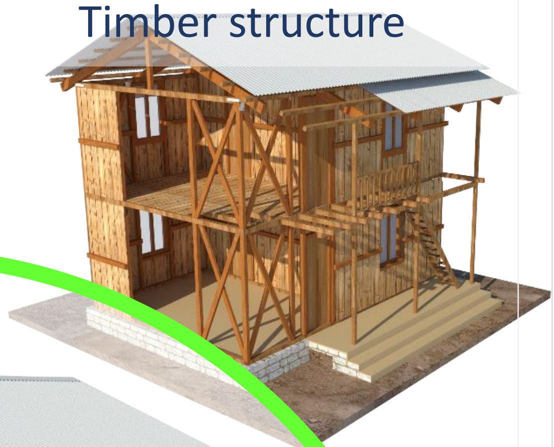
This manual is covering masonry structure in ground floor, and lightweight materials with timber or (steel) frame structure.

In case of masonry structure as construction method in ground floor , this manual is covering SMM(BMM) with RC band, SMM (BMM) with Wooden band, SMC with RC band and BMC with RC band.

G. Floor \ 1st Floor	SMM with RC band	SMM with wooden band	SMC with RC band	BMC with RC band	Timber	Steel
SMM (BMM) with RC band		×	×	×	○	○
SMM (BMM) with wooden band	×		×	×	○	○
SMC with RC band	×	×		○	○	○
BMC with RC band	×	×	△		○	○



Masonry structure

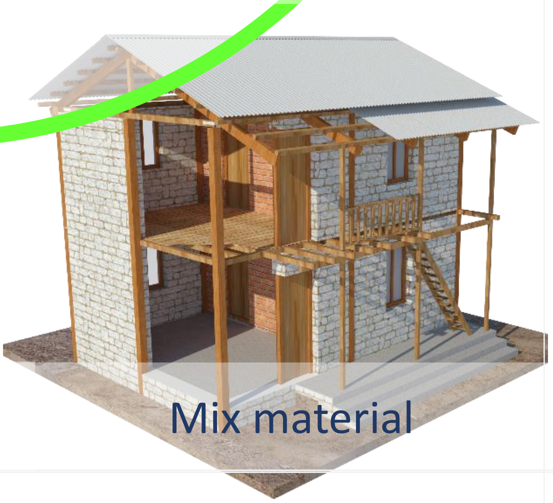


Timber structure

Hybrid structure

This manual is intended to cover only those buildings that are constructed using load bearing masonry structures in ground floor and timber framed with light weight wall.

Timber structure + infill brick wall



Mix material

The masonry structure at ground floor shall consist of all the earthquake resistant elements such as horizontal and vertical bands. R.C. or wood both can be used to construct these bands, but it shall be compliant with the respective minimum requirements.

The first floor shall be timber frame construction. Since, different construction technologies resulting from connection details and materials can be used to construct the timber framed structures, no any specific minimum requirements has been made till date. Hence, any construction details with reference to the provided specification shall be followed in first floor.

If first floor is constructed with steel frame structure, then strength of each structural items shall be equivalent to required strength mention in this manual.

Under reconstruction programme, if conditions of building are below, inspection shall be based on specification provided in this manual hence, structural calculation is not required.

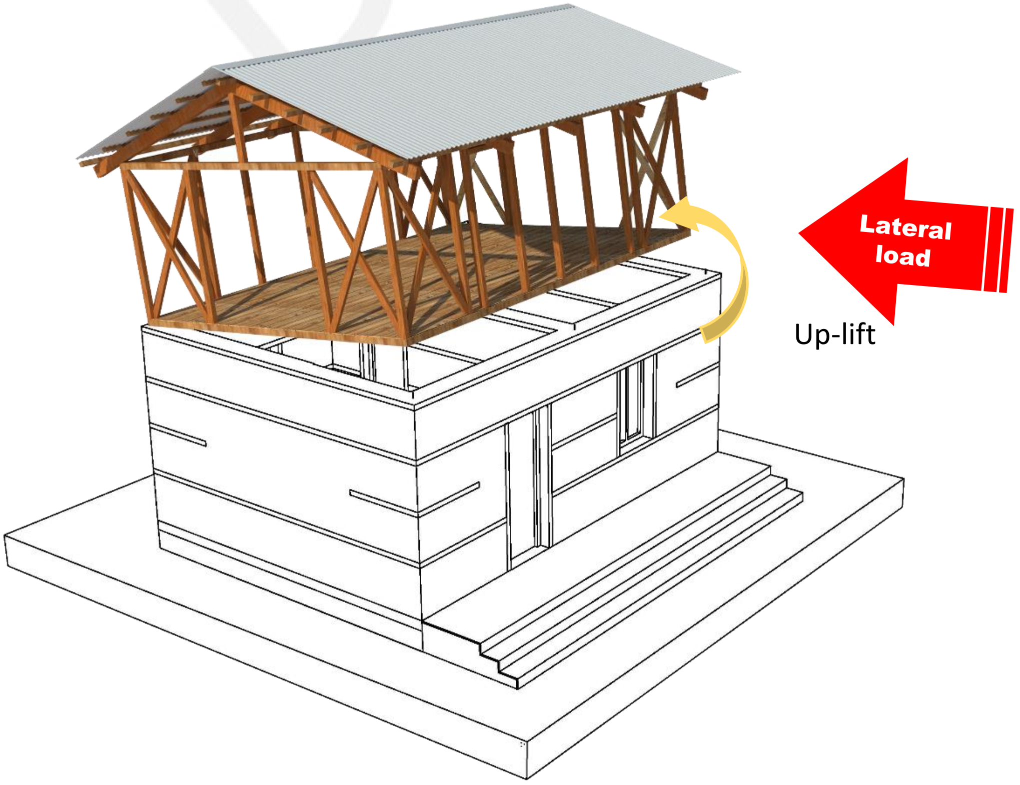
1. Upto two storey, ground floor is maonry structure, 1st floor is timber structure.
2. 1st floor area shall not be more than ground floor area.
3. Hight of building is less than 3m of ground floor, and 2.5m of 1st floor.

Failure Pattern

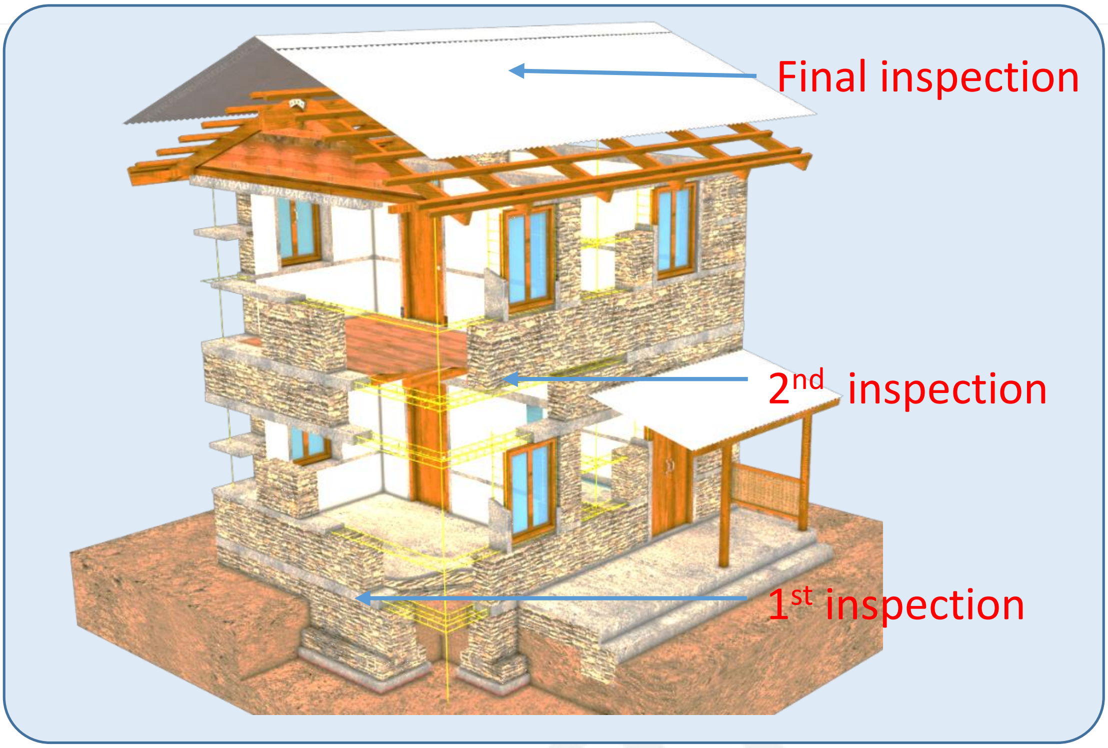
The horizontal ground motion action is similar to the effect of a horizontal force acting on the building, hence the term "Seismic Load" or "Lateral Load" is used. As the base of the building moves in an extremely complicated manner, inertia forces are created throughout the mass of the building and its contents. It is these reversible forces that cause the building to move and sustain damage or collapse.

1. If element of resisting against lateral loads is not enough. The building will be partially or totally collapse.

2. If there is poor connection between the ground floor and first floor with rigid structure, the building might happen uplift and rocking, or sliding behavior.



Timing of execution of inspection



1st inspection

The first inspection shall be done after completion of the construction up to plinth level. The appropriate inspection sheet according to the masonry typology mentioned in **annex 10** shall be used during the inspection. If the structure is found to be compliant then it can be certified for receiving 2nd tranche, else the correction order shall be given using the forms provided in **annex 11**.

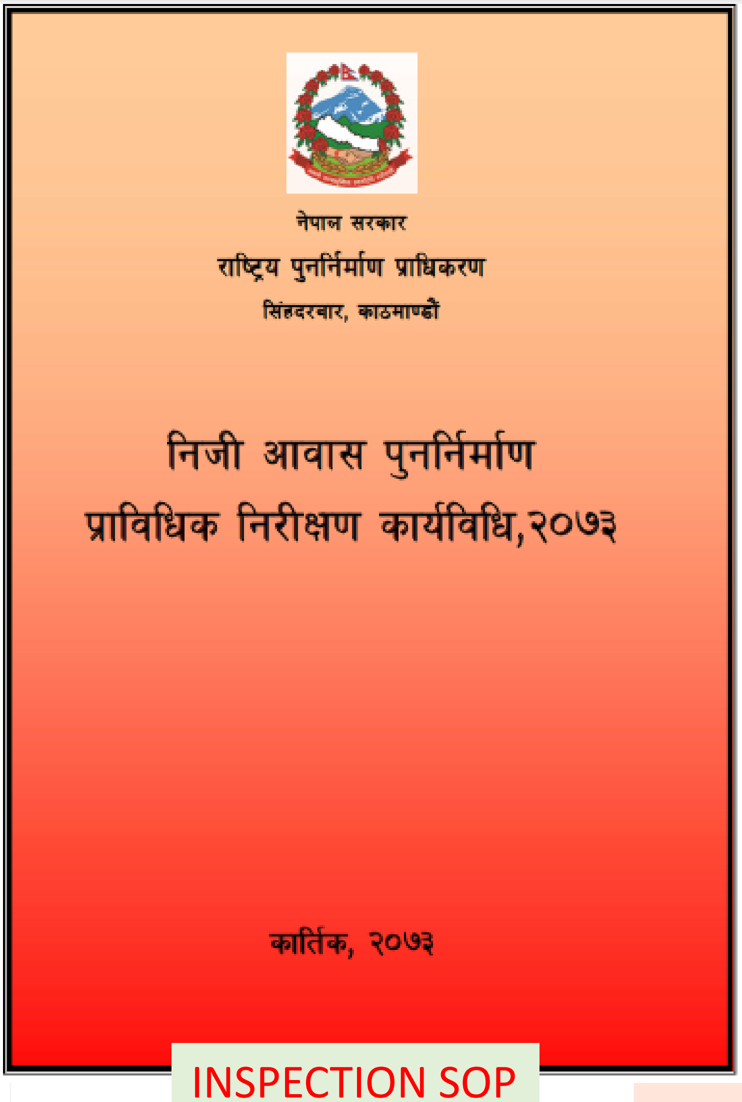
2nd inspection

The second inspection shall be done after completion of the roof band of one story and the first floor of the multistory house, the beneficiaries should apply for the inspection of the house and third installment using the form provided in **annex 6.** Technical Inspection Team should use the form in **Annex-13** to certify the house if the constructed house is according to earthquake resilient design and approved design.

If correction has to be made, **annex- 11** form shall be used by Technical assistance Team informing about the things to be correct.

Final inspection

The final inspection shall be done after completion of the roof. Technical Inspection Team should inspect and fill the form as specified in **Annex-15** and if the constructed house is found to be as approved design and earthquake resilient then it is recommended for "House reconstruction completion certificate"

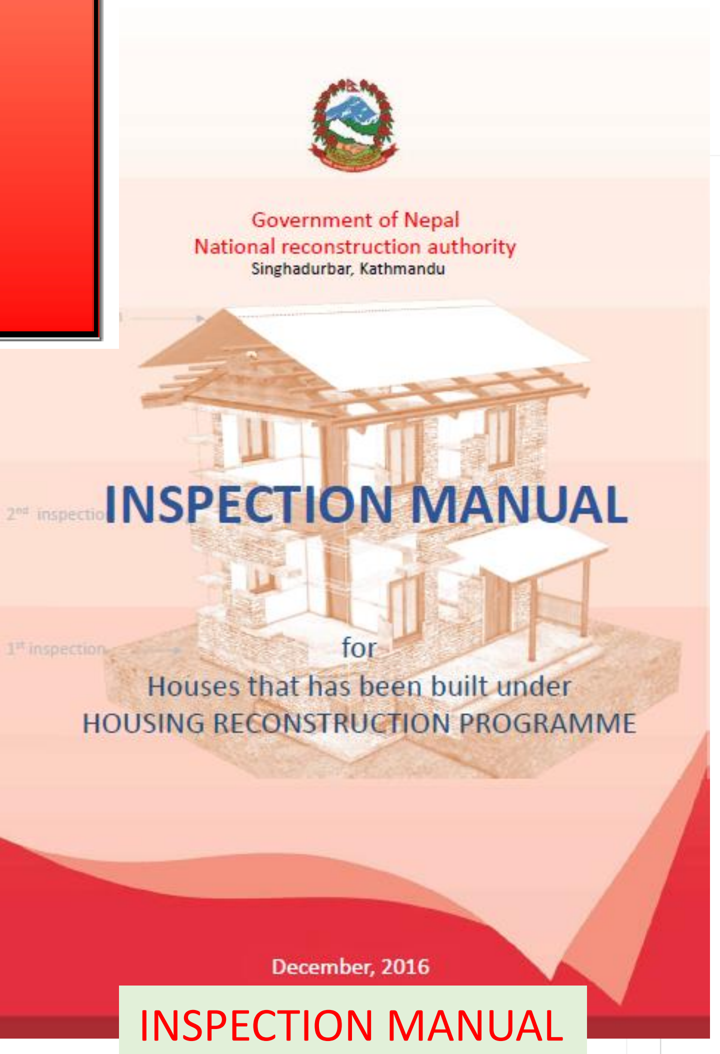


नेपाल सरकार
राष्ट्रिय पुनर्निर्माण प्राधिकरण
सिंहदरबार, काठमाण्डौं

निजी आवास पुनर्निर्माण
प्राविधिक निरीक्षण कार्यविधि,२०७३

कार्तिक, २०७३

INSPECTION SOP



Government of Nepal
National reconstruction authority
Singhadurbar, Kathmandu

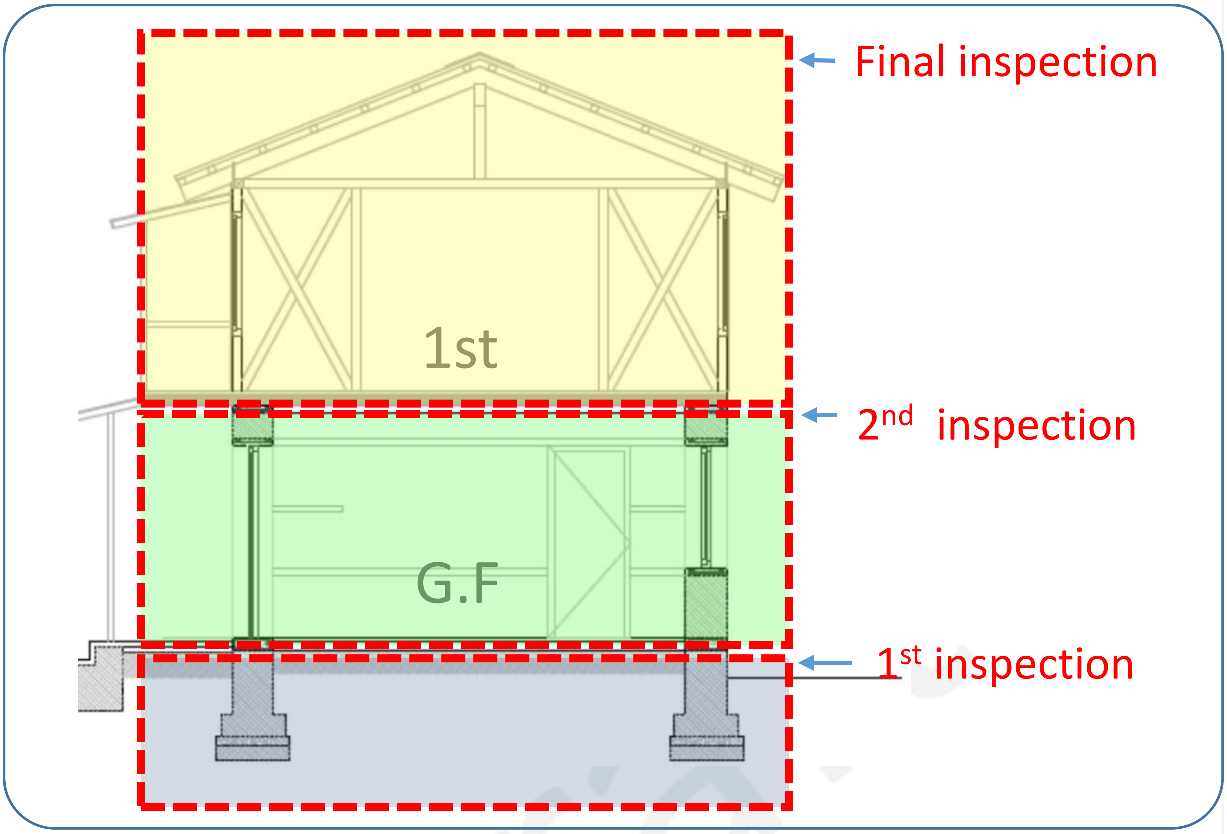
INSPECTION MANUAL

for
Houses that has been built under
HOUSING RECONSTRUCTION PROGRAMME

December, 2016

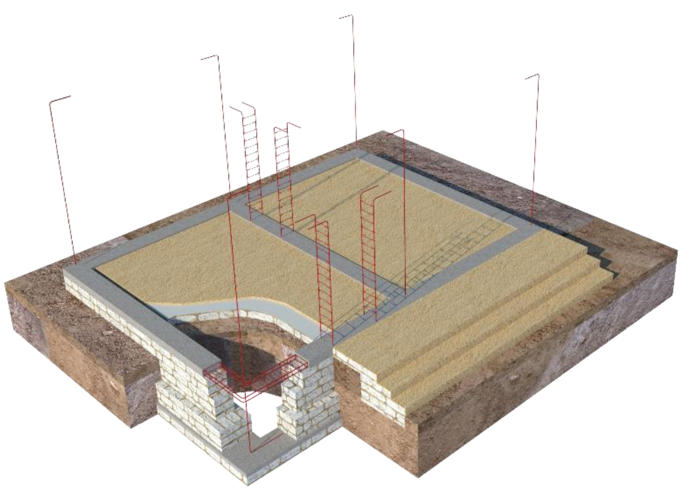
INSPECTION MANUAL

Timing of execution of inspection

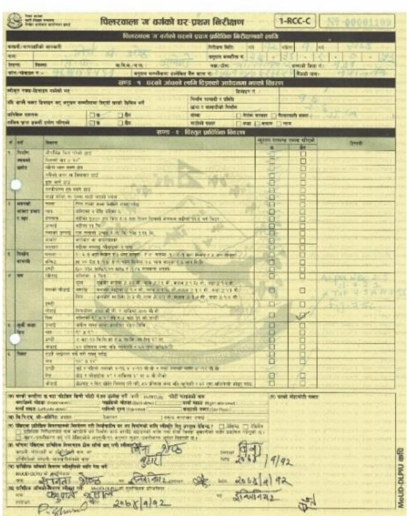


1st inspection

In order to carry out the first inspection of the hybrid structure, where the ground floor is constructed using load bearing masonry, the inspection shall be done on the basis of appropriate inspection forms (Annex 10) provided to inspect masonry building. Here, the structure shall be constructed on the basis of MRs. It shall be complaint to all the MRs or exceptional cases. If it is found to be non compliant correction order shall be given using the form provided in annex 11.



Level of construction for 1st inspection



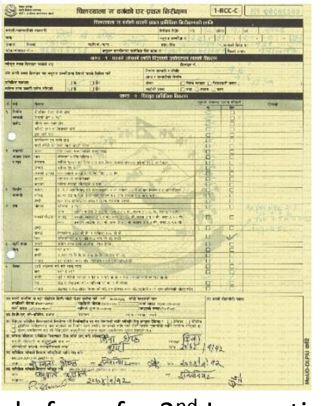
Sample from for 1st Inspection

2nd inspection

After completion of construction up to floor level, the 2nd inspection is carried out . Since, the ground floor is constructed using masonry structure, the forms that shall be used to inspect is same as the inspection forms used to inspect the masonry building (Annex 13) .

Here, the super structure shall have all the earthquake resistant features and constructed on the basis of MRs. If it is found to be non compliant correction order shall be given using the form provided in annex 11.

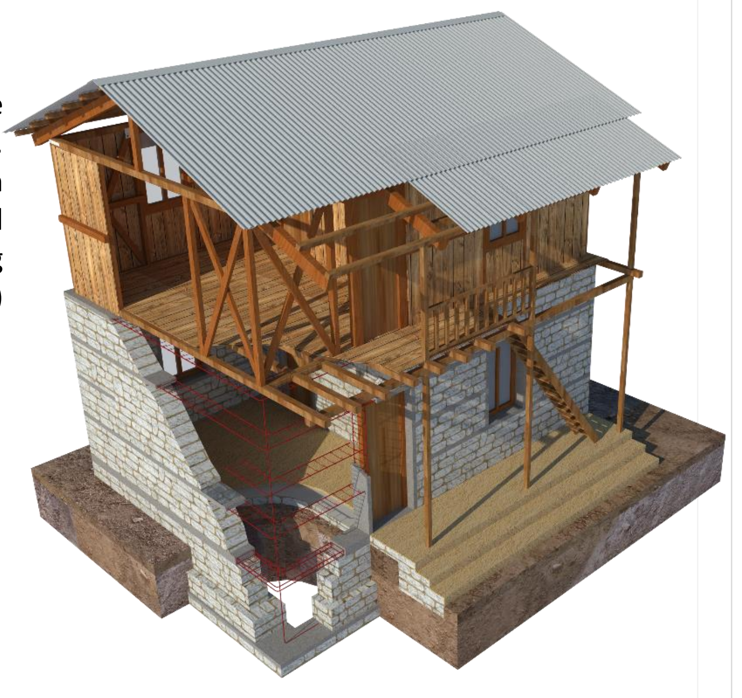
level of construction for 2nd inspection



Sample form for 2nd Inspection

Final inspection

The final inspection shall be done using the hybrid inspection forms. If all the description provided in the inspection sheets are found to be compliant then the building completion certificate (Annex 16) can be provided.



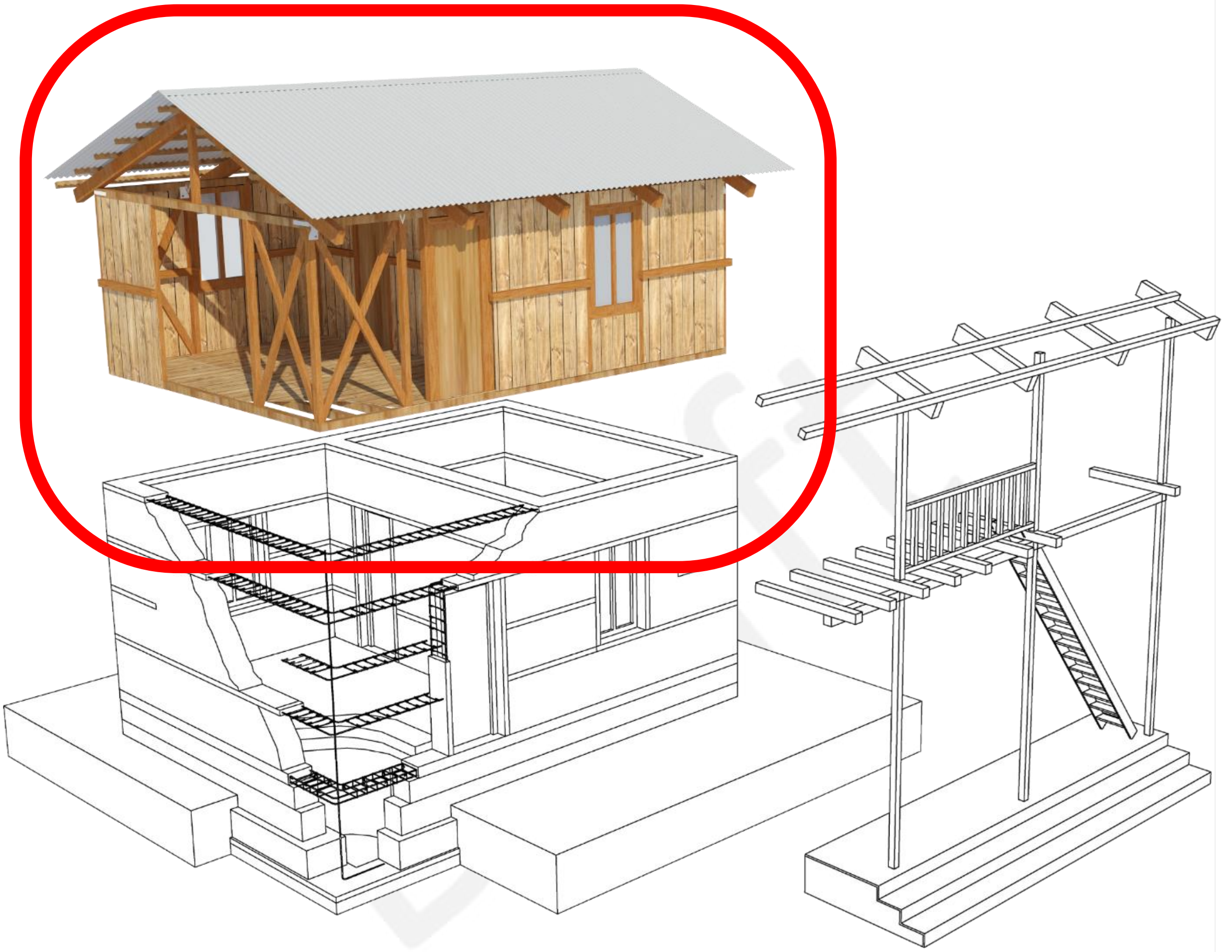
level of construction for 3rd inspection

Timing of execution of inspection

If 2nd inspection of masonry structure is satisfied, upper structure as hybrid can build.
Upper structure have to follow specification of this hybrid structure manual.

When upper part is complete, the final inspection shall be done using the hybrid inspection forms.
The main structural part of the building shall be inspected.

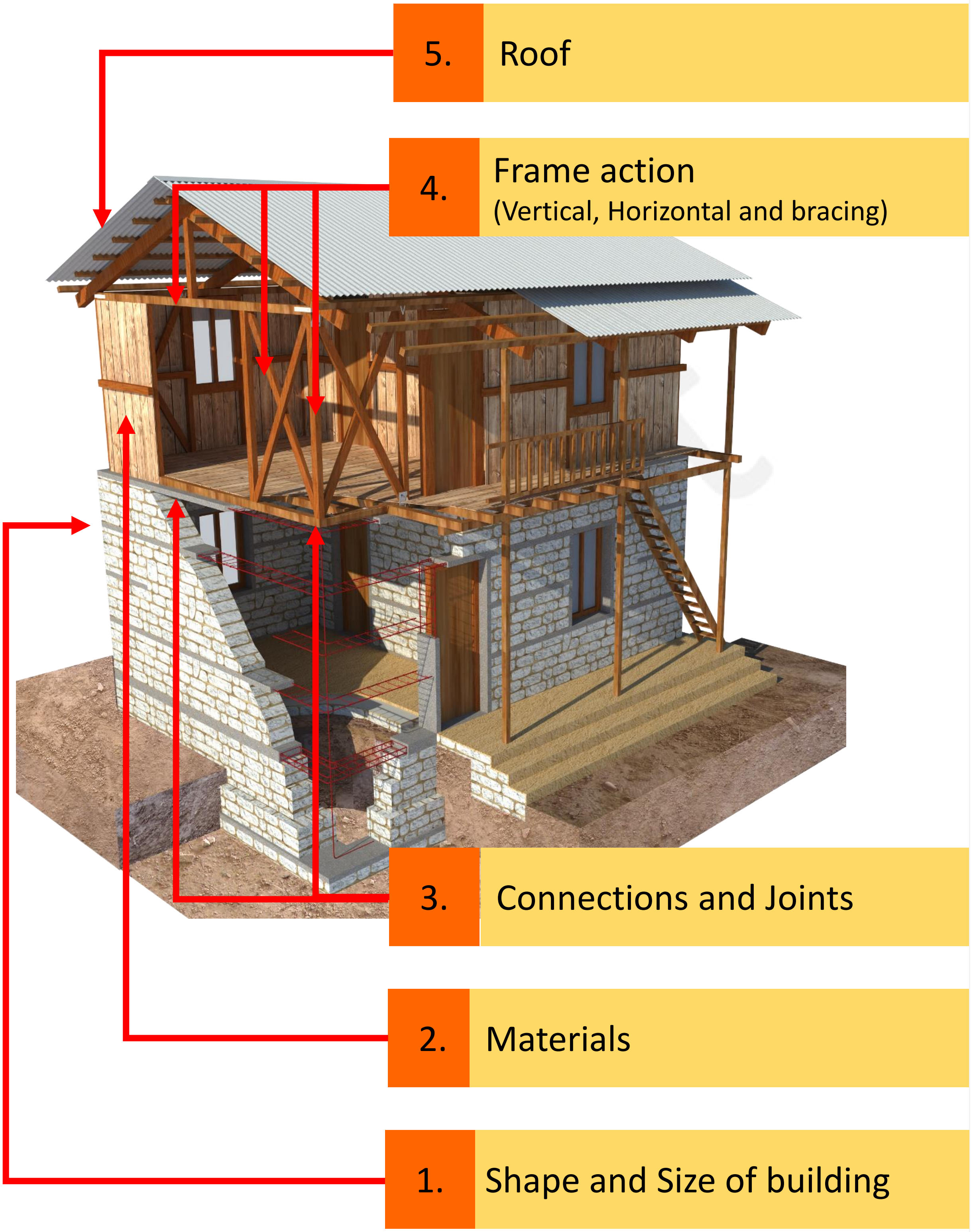
Main structural part for final Inspection



During the inspection of hybrid structures, only the main structural part of the building shall be inspected. Unless the projection of the building is within the Mrs, the inspection of the verandah can be ignored. Since, this manual has been prepared on the basis of the MRs of masonry structures hence, if the projection (Verandah) in the first floor exceeds the MRs then the detail structural calculation shall be done separately to strengthen these areas.

PART-2: Technical Specification of Hybrid Structure

Key Items of final inspection of hybrid structure



Inspection items of hybrid structure

1. Shape and Size of building

Simple rectangular shapes behave better in an earthquake than shapes with projections. The inertia forces are proportional to the mass (or weight) of the building and only building elements or contents that possess mass will give rise to seismic forces on the building.

2. Materials

Inadequate materials doesn't have sufficient stability and strength to withstand the lateral forces. Hence, use of these substandard materials might leads to the failure or ultimately collapse of the overall structure.

3. Connections and Joints

If there is poor connection between the ground floor and first floor with rigid structure, the building might happen uplift and rocking behavior, when the lateral load is imposed on to the structure.

4. Frame (vertical, Horizontal and Bracing)

Earthquake-induced inertia forces will be distributed to wall framing consisting of bottom and top plates with diagonal braces. Therefore, frame should be support each other as horizontally and vertically.

Wall framing consisting of bottom and top plates should have diagonal braces, or sheathing boards so that the frame acts as a shear or bracing wall.

Horizontal diagonal braces are used to resist the frame against lateral loads due to earthquake and wind.

5. Roof

In order to resist against lateral forces, proper connection of roof to the vertical post and top plate shall be done. Depending upon the structures cross bracing is also required.

1. Shape and Size of building

Requirements

No.	Category	Sub Category	Description	
1.	Shape and Size of building	No.of storey	Not more than two storey	
		Shape of house, Span of wall	1st Floor	Regular shape. The wall line of upper storey should be on the wall of lower storey. The wall line should not cantilever. Therefore, the span of wall also same as lower storey.
		Height of wall	1st Floor	The wall height shall not be more than 2.5m.

Why important?

No. of storey: The seismic load is represented by distinctly different from dead load, and live load. The inertia forces are proportional to the mass (or weight) of the building and only building elements or contents that possess mass will give rise to seismic forces on the building. Therefore, the lighter the material, the smaller will be the seismic force. If attic is used as storage, heavy mass is on the top of building, hence, seismic force become high.

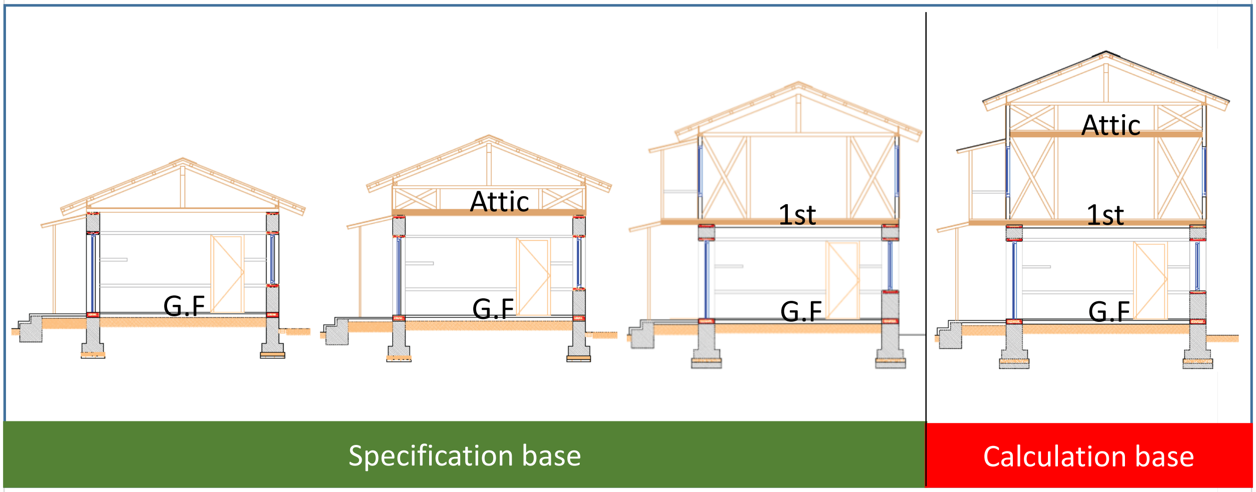
Shape and Size of building: Simple rectangular shapes behave better in an earthquake than shapes with projections. Torsional effects of ground motion are pronounced in long narrow rectangular blocks.

Exception

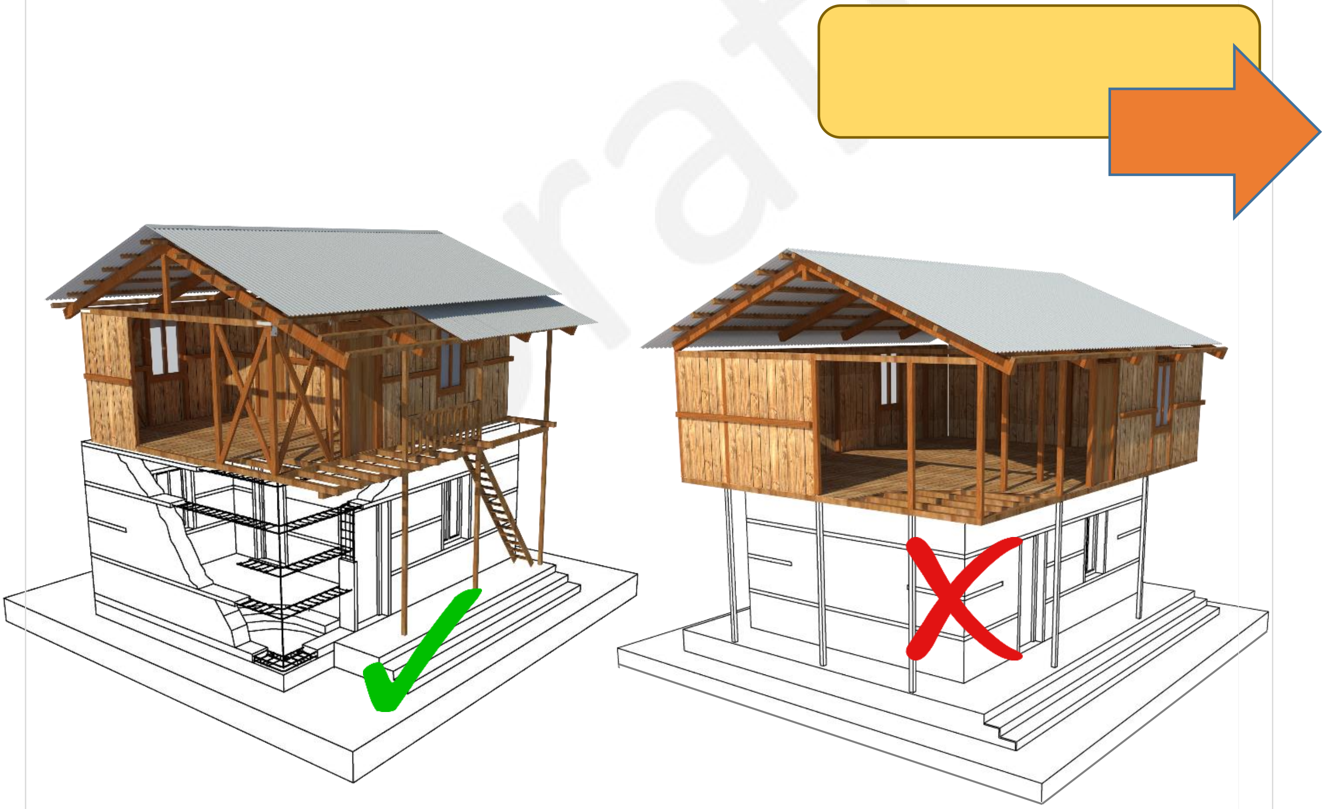
- If structure is found to be safe after structural calculation.
- In case of two storeys plus attic constructed by stone masonry in mud mortar in ground floor, and timber in first floor and attic, or wall height is more than 2.5m, if structure is found to be safe after wall ratio (quantity of brace) calculation. Simplified calculation method is shown in P00.

Inspection methodology

It can be checked by the observation and measurement.



- Unto two storey, inspection is specification base, however, if upper storey has attic, structural calculation is necessary.



- The wall line of the upper storey should be supported by bearing wall line of the lower storey.
- The column should rest on the bearing wall of the lower storey. No cantilevered storey elevation.

2. Materials

Requirements

No.	Category	Sub-category	Description
2.	Materials	Nail	Common wire nails shall be made of mild steel having a minimum tensile strength of 550N/mm2. Nails with appropriate diameter and length shall be provided.
		Bolt	It shall be used in such the number, diameter, length, spacing as each design/specification.
		Metal Plate	It shall be used in such the number, diameter, length, spacing as each design/specification.
		Rebar	High strength deformed bars with fy = 415 Mpa /500 Mpa.
		Timber	Treated and well seasoned hard wood or locally available wood without knots shall be used..

Why important?

- Inadequate materials doesn't have sufficient stability and strength to withstand the lateral forces. Hence, use of these substandard materials might leads to the failure or ultimately collapse of the overall structure.
- Moisture can cause wooden surfaces to swell and deform. Excessive moisture will lead the wood to decay, caused by decay fungi that ruin the material completely.
- Shrinkage of wood on drying is relatively large. Joint loosen easily due to contraction in the direction perpendicular to fibers. Therefore dry wood shall be used with moisture content less than 20 %.
- Wood can decay from repeated change of moistures. Therefore seasoned wood should be used in construction.

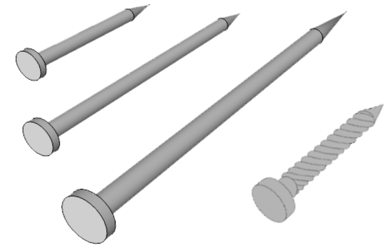
Inspection methodology

It can be checked by the observation and measurement. Thickness of the steel plate shall be adequate and it can be checked by measurement.

2.1 Nail

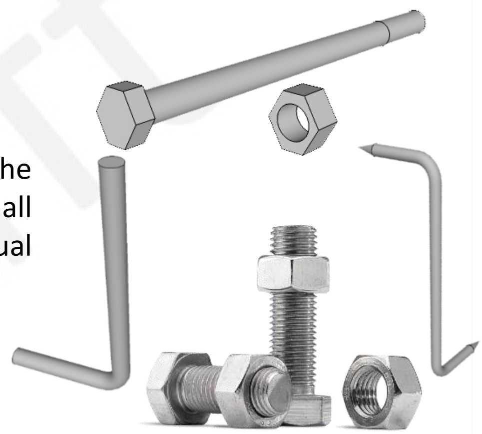
The things that need to be checked in nail are as follows:

a) Nails diameter:
Nail diameter shall be in between 1/11 and 1/6 of the least thickest of the members to be connected.
b) Nails length:
The length of a nail shall be at least 2.5 times the thickness of the thinnest member and it shall penetrate the thicker member by 1.5 times the thickness of the thinner member, whichever is further.
c) Number of nails:
The number of nails in a group should not exceed 10 in one rows in the direction of the force.



2.2 Bolts:

- When a number of bolts are used in a joint, the allowable load in withdrawal or lateral resistance shall be the sum of the allowable loads for the individual bolts.



2.3 Metal Plate:

- The bolts shall be arranged in such the size, thickness, spacing as each design/specification.

2.4 Rebar:

- Heavily rusted rebar should not be used.
- After rubbing the steel bar, if stain is present on fingers, but if the flakes doesn’t come off then the rust is acceptable.
- The thickness of rebar is checked by using vernier caliper. Ductility of rebar can be checked by bending it at 90° and if small cracks are found ductility is insufficient.

2. Materials

2.1 Wood

Exception

Tolerances:

- Permissible tolerances in measurements of cut sizes of structural timber shall be as follows:

a) For width and thickness:

1)	Up to and including 100mm	+3mm -0mm
2)	Above 100mm	+6mm -3mm

b) For length: +10mm -0mm

Inspection methodology

- Timber treatment can be identified by the observation or questionnaires survey with the house owner and mason.
- Typology of the wood can be identified by observation and field test.
- Defects in timber can be identified by observation.
- Moisture content in the timber can be identified by oven-dry method.

Wood can readily be identified as a hardwood or softwood by the following procedure:

- The color of hardwood is dark brown and light brown in softwood.
- When the thumb nail is pressed against hardwood it will not leave a mark but when it is pressed in softwood and pull it along a surface it leaves a scratch mark. Deeper the mark, the softer the wood.

When pressed by tip of nail leave a mark (Soft wood)

When pressed by tip of nail doesn't leave a mark (Hard wood)

Source: http://www.instructables.com/id/Hard-Wood-or-Soft-Wood%3F/

- Classification of Woods According to Characteristics
- Structural properties and Safe working stress

List of Hardwood and softwood

HARD WOOD		SOFT WOOD	
Babul	Mesua	Chir	Simal
Blacksiris	Oak	Deodar	Uttis (Red)
Dhaman	Sain	Jack	Uttis (White)
Indian Rose Wood (Shisam)	Sal	Mango	
Jaman	Sandan	Salla	
Sissao	Teak		
Khair			

Source: NBC 203:2015

- Timber treatment
  It such as use of coal tar or any other preservative can prevent timber from being decayed and attacked by insects. Timber shall not have defects such as dead knot, decay or rot, discoloration

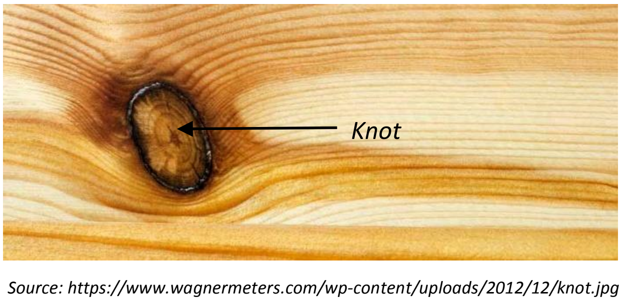
- Moisture content in Timber:
  Moisture content means the weight of water contained in wood, expressed as a percentage of its oven dry weight. It can be determined by the oven-dry method.

S.N	Kinds of wood	Weight (12% moisture content)
1	SAL (AGRAKH)	56
2	SISAU	50
3	KHOTE SALLA	33
4	GOBRE SALLA	32
5	UTTIS (RED)	36
6	UTTIS (WHITE)	34
7	CHAMP	33
8	SATISAL	38
9	ASNA	46
10	PHALAT	60
11	TOONI	37
12	SEMAL	25
13	OKHAR	45
14	OAK	64
15	KHAIR	60
16	BIJYASAL	49

Source: NBC 112:1994

Defects in Timber:

- Dead Knot: It is the knot in which the layers of annual growth are not completely intergrown with those of the adjacent wood. It is surrounded by pitch or bark. The encasement may be partial or complete.



Source: https://www.wagnermeters.com/wp-content/uploads/2012/12/knot.jpg

3. Connections and Joints

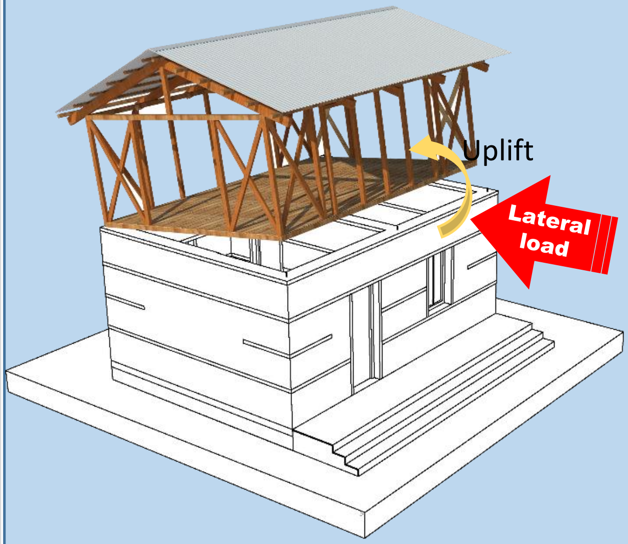
Requirements

No.	Category		Description
3.	Connections and joints	Connections between lower and upper structure	It should be arranged properly for connecting ground floor and first floor.
		Joints of structural member	All the structural members (frame action) shall be properly connected by nails or bolts such as drawing / specification.

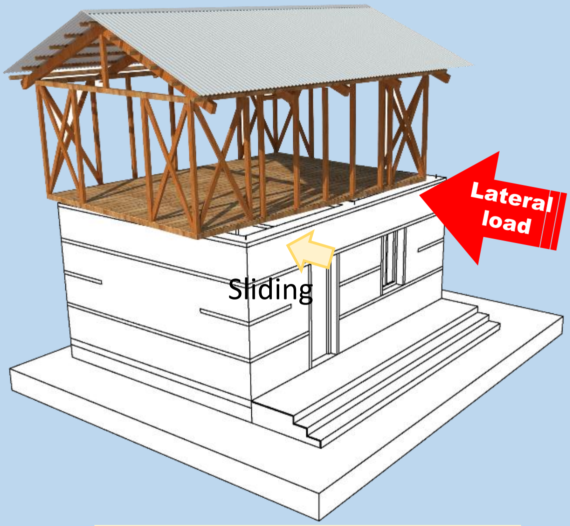
Why important ?

Connections between lower and upper structure

- If there is poor connection between the ground floor and first floor with rigid structure, the building might happen uplift and rocking behavior, when the lateral load is imposed on to the structure.
- Sliding of the first floor as a whole is sometimes seen when there are no anchor bolts connecting the ground floor and the first floor.
- Stone masonry in mud mortar with wooden band, bonding strength is very poor, therefore, uplifting behavior will be happen easily.



Uplifting or Rocking



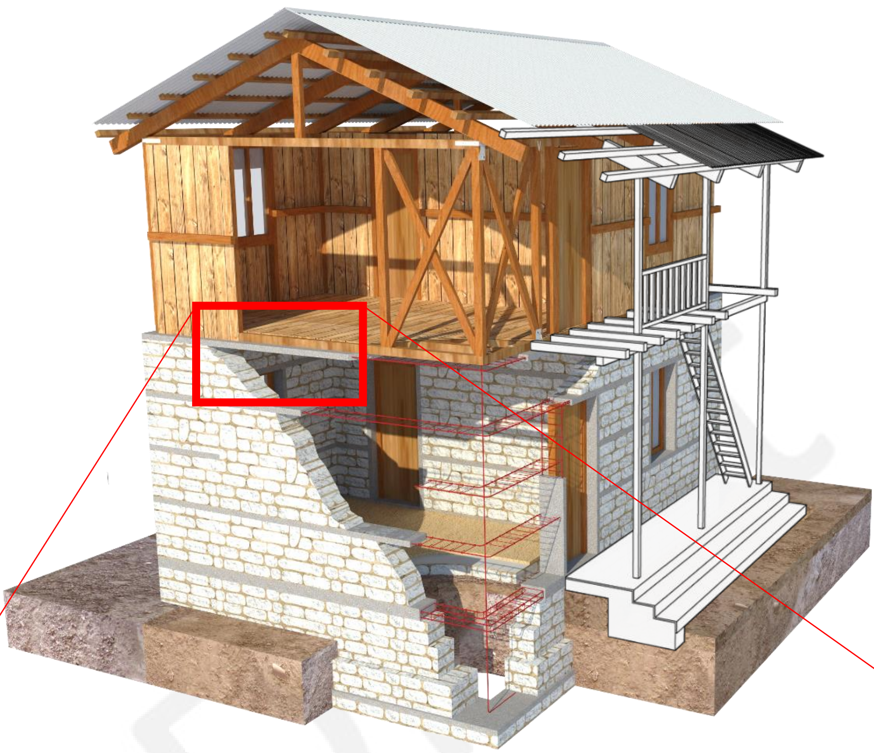
Sliding

Joints of structural member

- The failure of the joints connecting structural member such as vertical, horizontal and bracing are frequently occurs. Structure member should be uniform, so that the framing acts as earthquake resistance elements.

Technical Specification

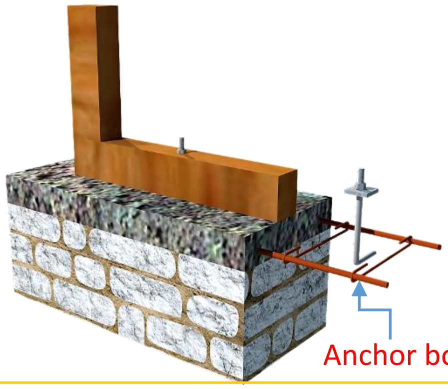
In case of floor band is made by Reinforcement Concrete.



Connection of floor RC bands with wooden base plate

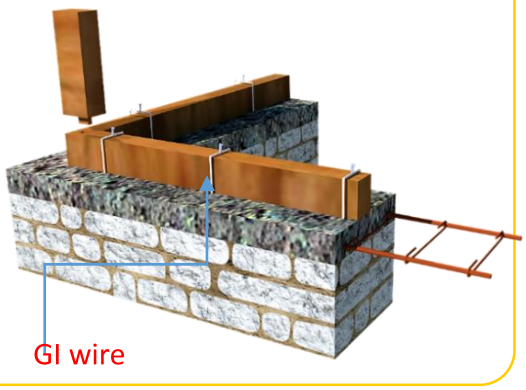
Option 1

Connecting base plate and floor bands by anchor bolts whose minimum diameter is 12mm and length 350mm. The anchor bolt are installed at the minimum spacing is 2meter.



Option 2

Connecting base plate and floor bands by GI wire 3.25mm (10 Gauge) at the spacing of 450mm c/c throughout the wall.



Technical Specification

In case of Stone masonry in mud mortar with wooden band

Option 1

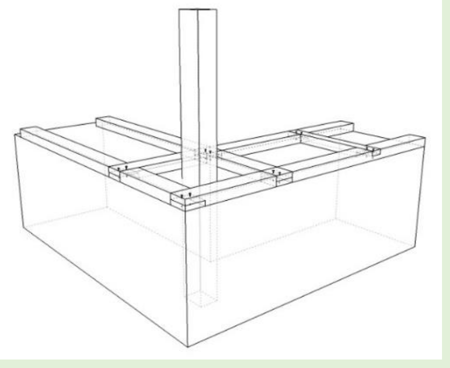
The vertical member is continuous from ground floor to 1st floor.

Option 2

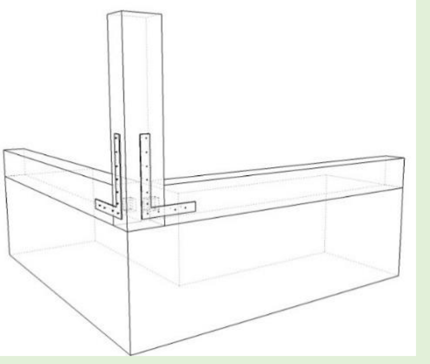
Wooden floor band and lintel band should be connected.

Technical Specification

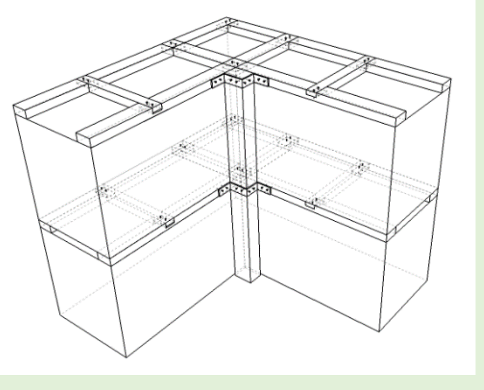
Wooden vertical member should be properly connected to horizontal member as shown in figure.



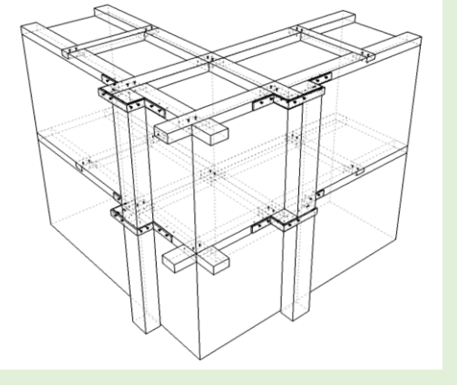
Continue vertical member from lower structure (masonry)



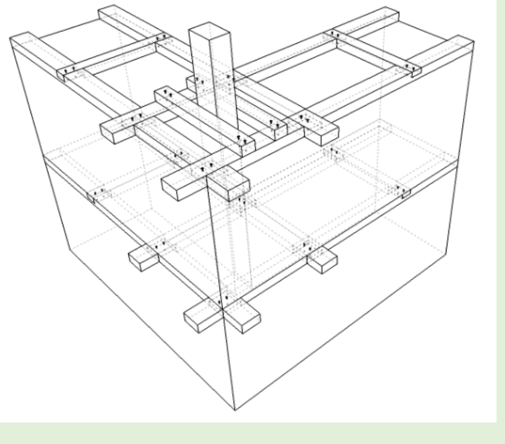
Properly connect to horizontal member

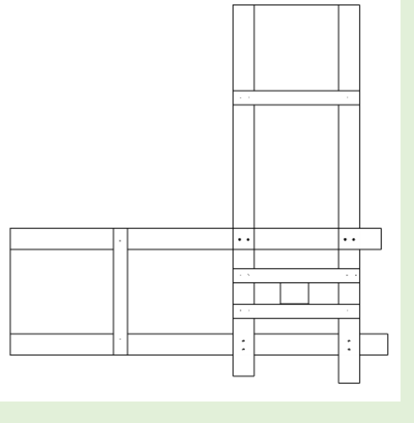


Vertical Member -Inside



Vertical Member -Outside



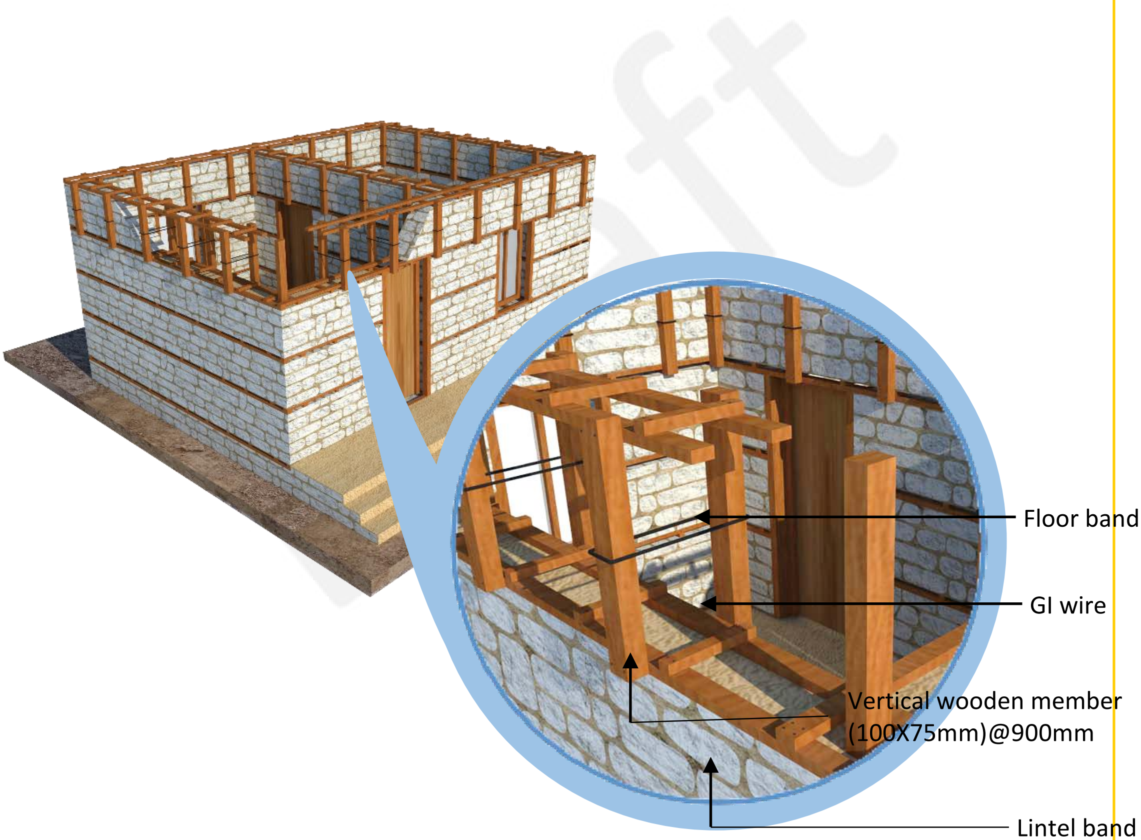


Technical Specification

In case of Stone masonry in mud mortar with wooden band

If vertical member is continuous from ground floor, connecting wooden floor band and lintel band is not necessary.

Option 2-1: Connecting wooden floor band and lintel band with vertical wooden member.



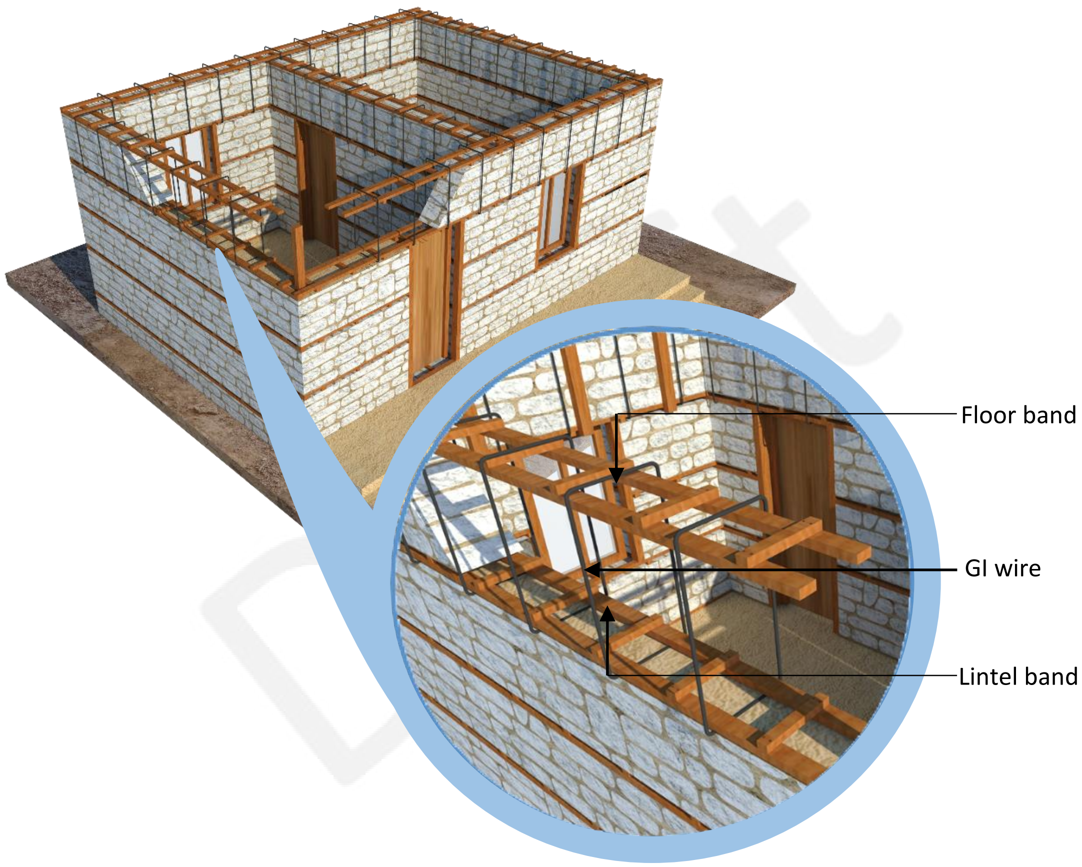
Steps:

1. Connect lintel band and floor band with vertical wooden member on both side of the wall.
2. Place the vertical wooden member (100X75mm)@ 900mm c/c on both side of the wall.
3. Tie both vertical member with GI wire

Wooden floor band and lintel band should be connected.

Technical Specification

Option 2-2: Connecting wooden lintel band and lintel band with GI wire



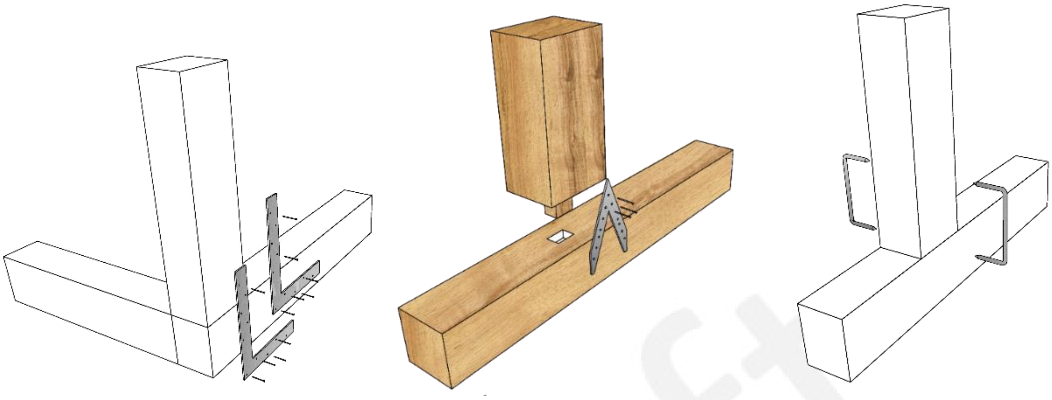
Steps:

1. Insert and tie the GI wire 3.25mm (10 Gauge) at the spacing of 450mm c/c. starting from lintel to the floor band through out the wall.

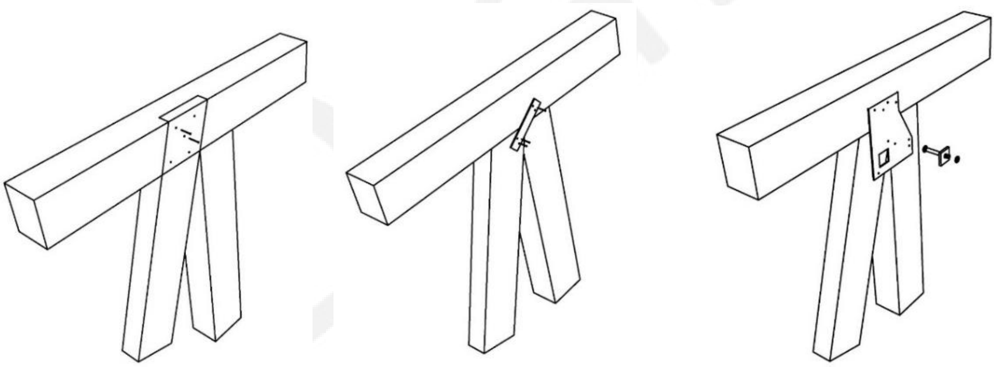
Joints in Wood Frames

Technical Specification

Connection between the wooden member



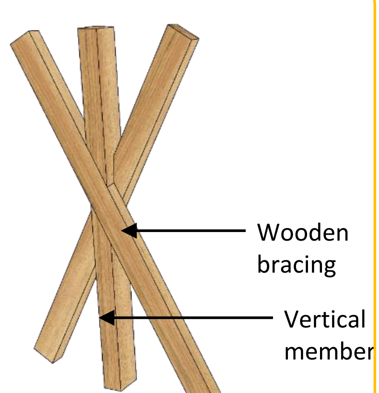
Connection detail of fixing vertical post and base plates



Connection between brace and horizontal/ vertical member

Inspection procedure

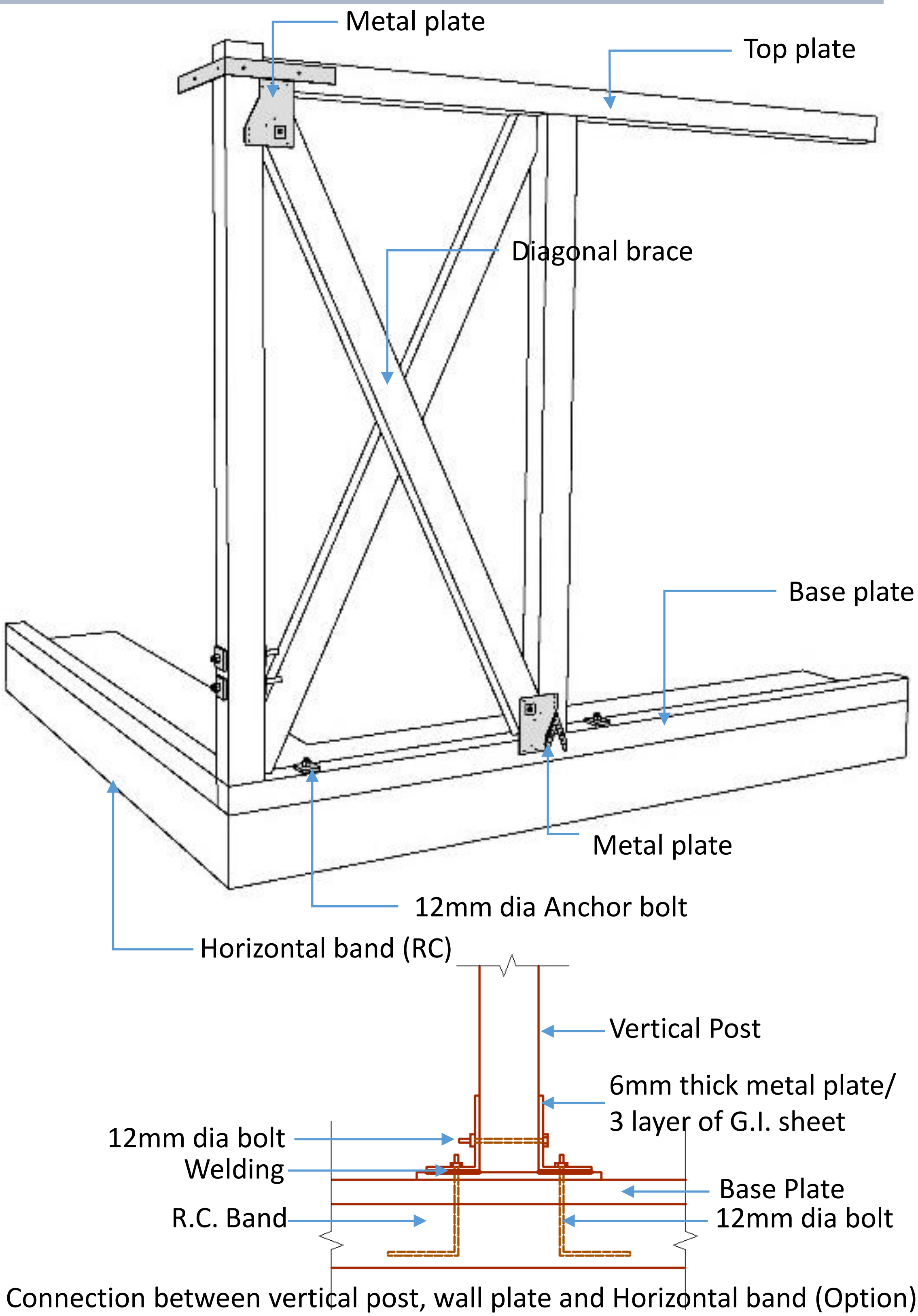
- The connections that needs to be checked are:
  ✓Connection between wooden member
  ✓Connection between wall plate and floor band.
  ✓Connection of bracing with vertical member and wall plate.
  ✓Connection between wooden lintel and floor band.



Connection between two wooden braces and vertical member

Technical Specification

Connection detail using metal plate



Connection between vertical post, wall plate and Horizontal band (Option)

4. Frame Action [Vertical, Horizontal and Brace]

Requirements

No.	Category	Description		
4.	Frame	Each member shall be properly connected.		
		Vertical member	It should continue to base plate to top plate.	
			Size	Main column shall be more than 75x75(mm) in hard wood, 100x75(mm) in soft wood.
			Spacing	It shall be less than 1200mm.
		Horizontal member	It shall be same level and continuous	
			Base plate/ Top plate	Size of wood shall be more than 100x75mm
		Bracing	location	It should locate at each corner and balanced on plan
			Direction	It should be balanced (symmetrical) on wall line.
			Size/number	It shall be as per design / specification.

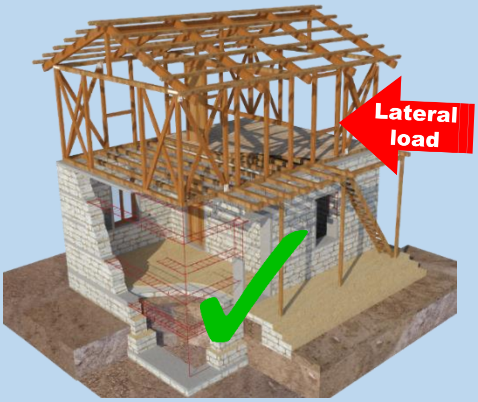
Exception

- If using steel frame the strength of each member should be equivalent to the strength of wood.
- If structure is found to be safe after structural calculation.
- Size of vertical and horizontal member depend on various span. Refer to NBC 203 and 204.

Why important?

Earthquake-induced inertia forces will be distributed to wall framing consisting of bottom and top plates with diagonal braces. Therefore, frame should be support each other as horizontally and vertically.

Wall framing consisting of bottom and top plates should have diagonal braces, or sheathing boards so that the frame acts as a shear or bracing wall.



Diagonal braces are main element of resisting against lateral loads due to earthquake and wind.

Fundamental items

1. Each member (vertical, horizontal and brace shall be properly connected.
2. Vertical member shall continue to top of beam.
3. Vertical member shall be placed at proper spacing.
4. Horizontal member shall be continuous in same level.
5. Brace shall be symmetrical and located at each corners.
6. Brace shall be provided from floor to top.

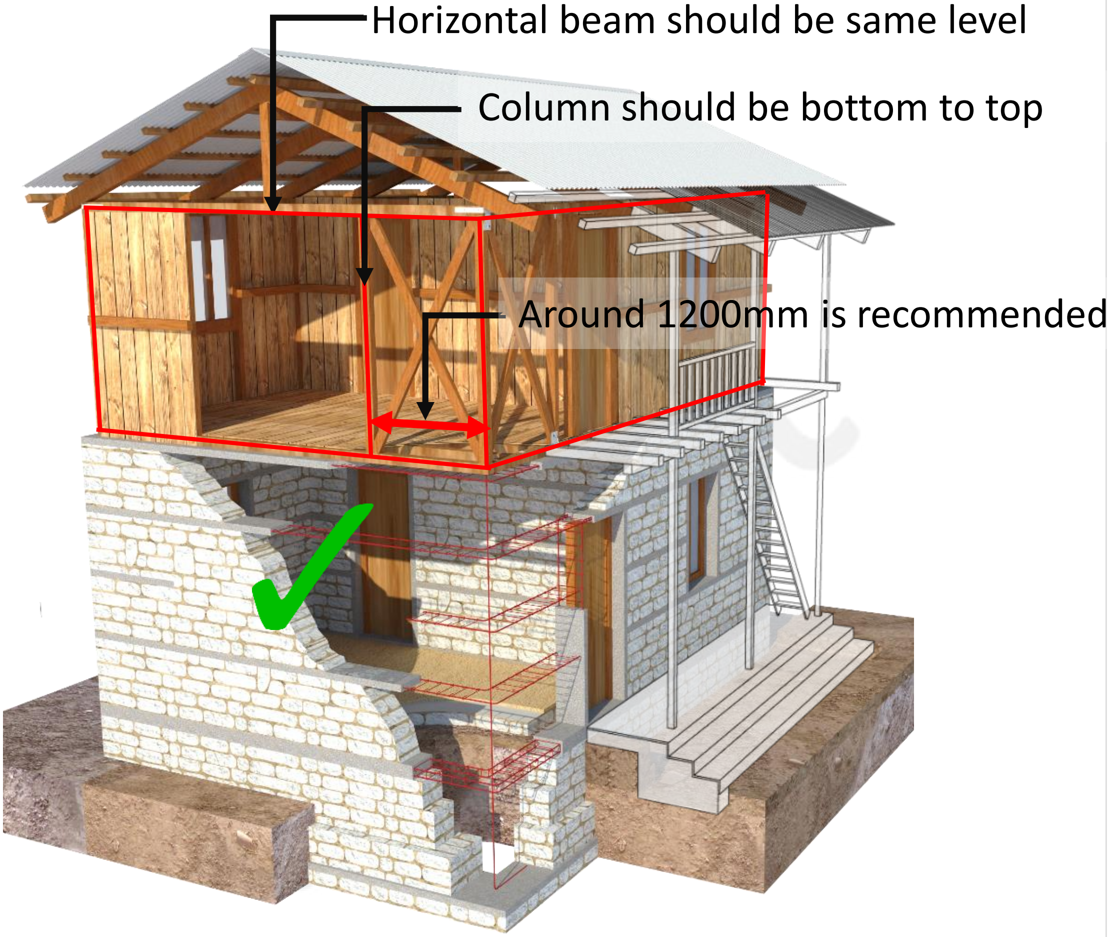
Case 1.

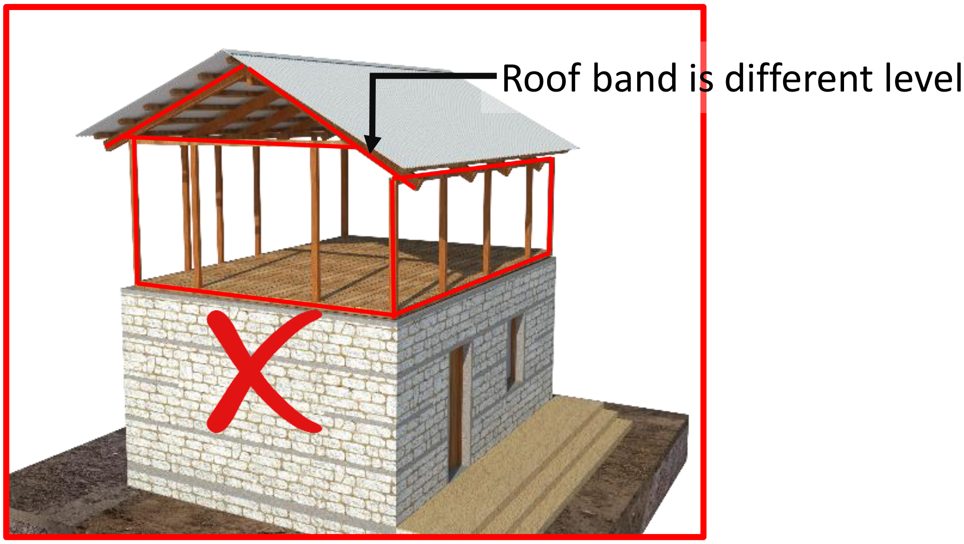
Case 2.

Vertical and Horizontal member

1. Standard type

Technical Specification





2. Traditional type

Technical Specification

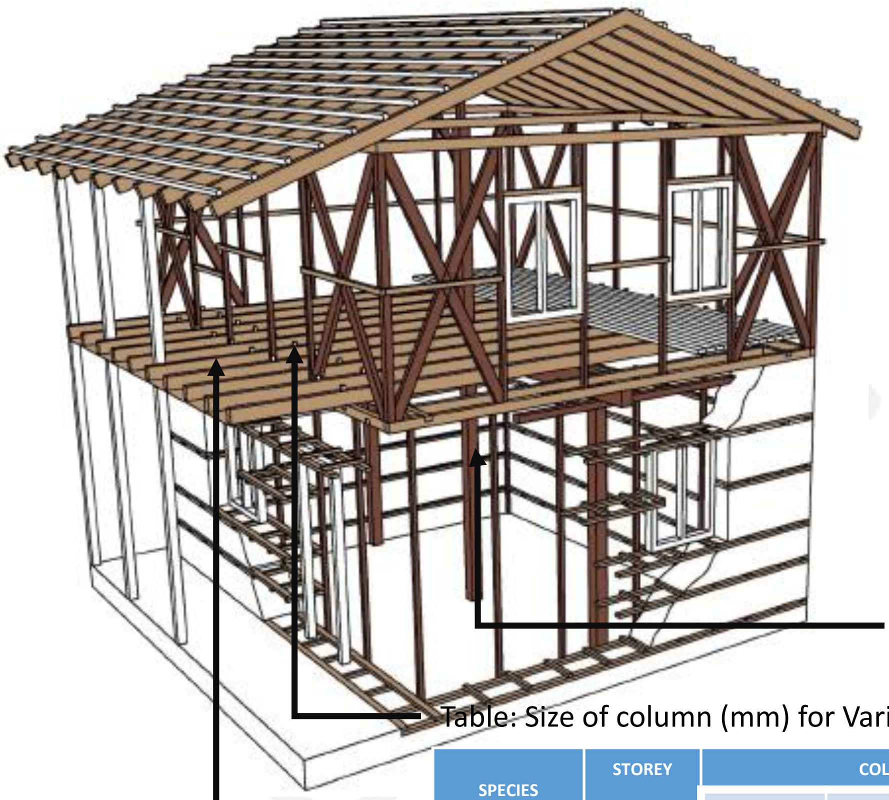


Table: Size of column (mm) for Various Spans

SPECIES	STOREY	COLUMN SPACING			
		2m	2.5m	3m	3.5m
Hard Wood	Ground	110x110	110x110	120x120	130x130
	1st	90x90	90x90	100x100	100x100
Soft Wood	Ground	120x120	120x120	130x130	140x140
	1st	90x90	100x100	100x100	110x110

Table: Depth of Beams (mm) for Various Spans

SPECIES	SPAN				
	2m	2 to 2.5m	2.5 to 3m	3 to 3.5m	3.5 to 4m
Hard Wood	190	220	240	270	300
Soft Wood	230	270	310	340	370

*For hybrid structure, if column span is 1200mm, column size is 75x75(mm) in hard wood, 100x75(mm) in soft wood.
However, the minimum dimensions of members for different span shall be as tabulated in Table.

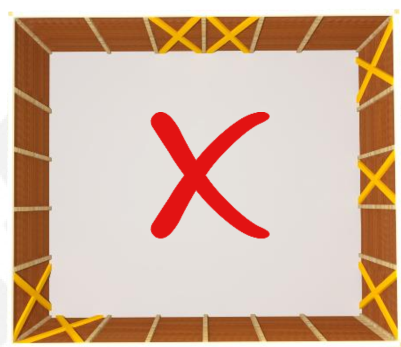
Bracing

Technical Specification

Location

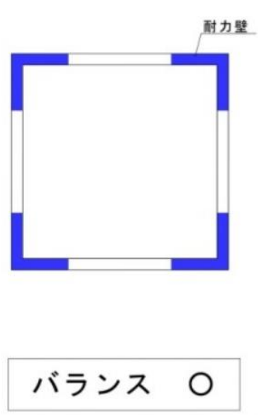
It should locate at each corner and balanced on plan

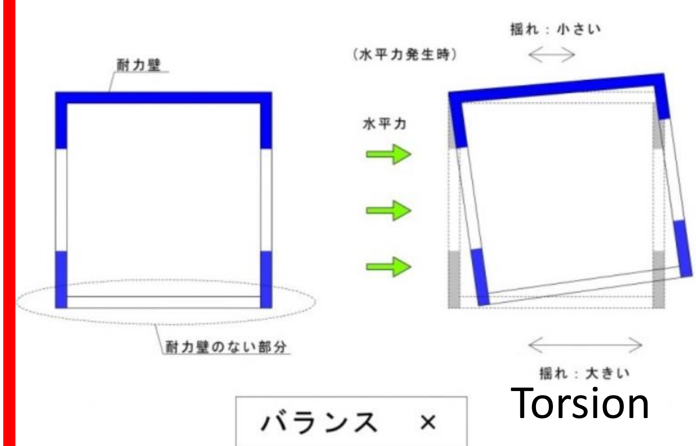
Balanced



Un-Balanced

Diagonal brace should be located each wall and corner. And way of brace member also should be balanced, shown in figure.



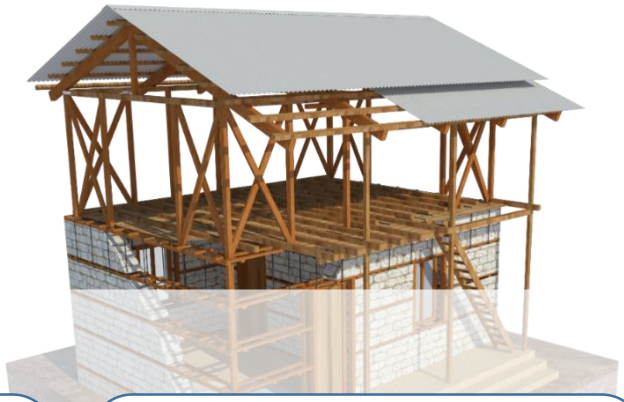


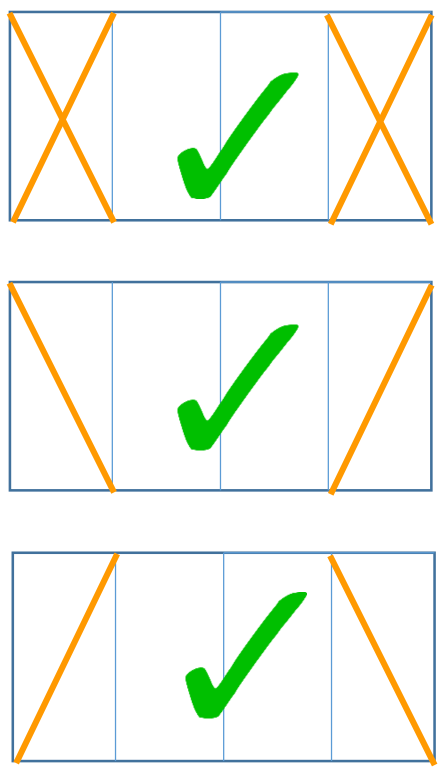
Technical Specification

Direction

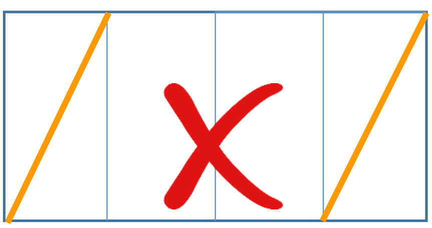
It should be balanced (symmetrical) on wall line.

Achieve adequate seismic resistance, provide diagonal bracing members in the planes of walls as well as horizontally at from the bottom to the top of walls as shown in Fig.

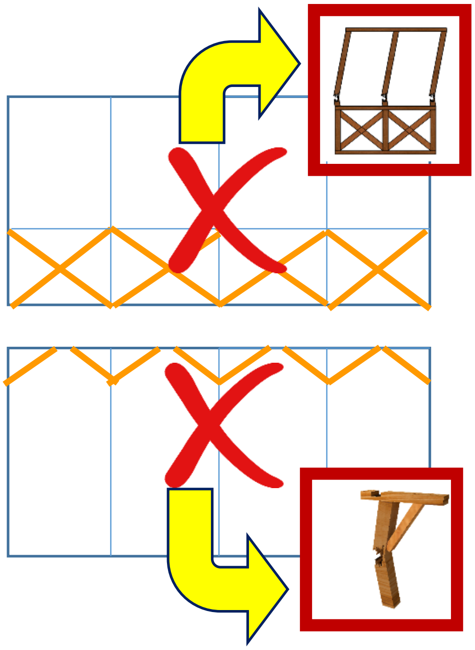




Brace should be Balanced



Direction is wrong way.
It shall be balanced.

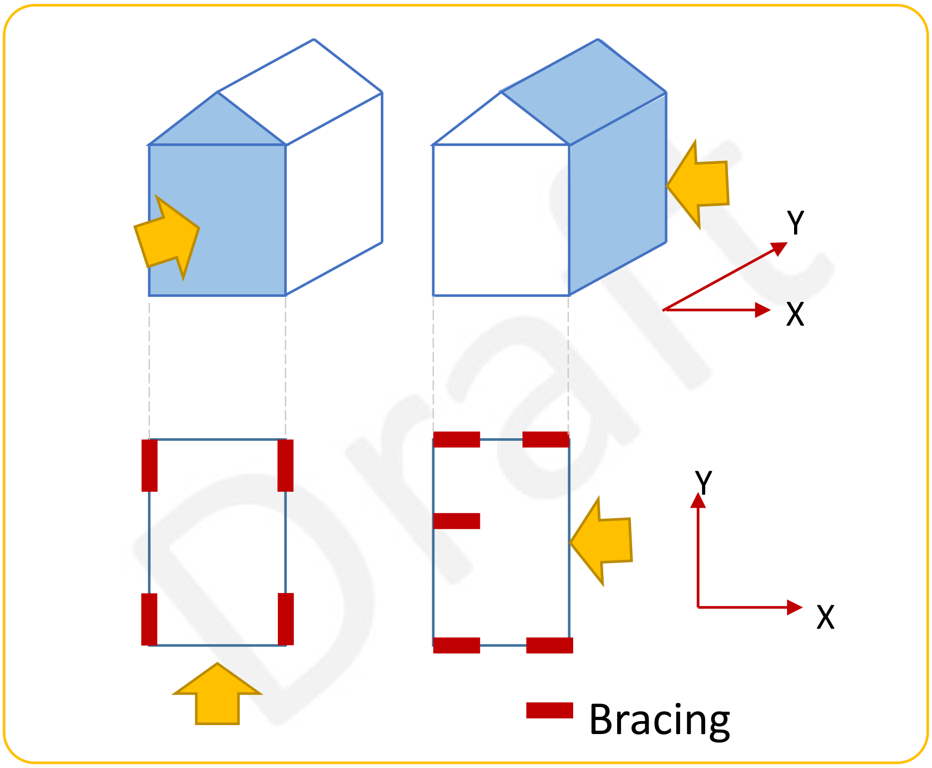


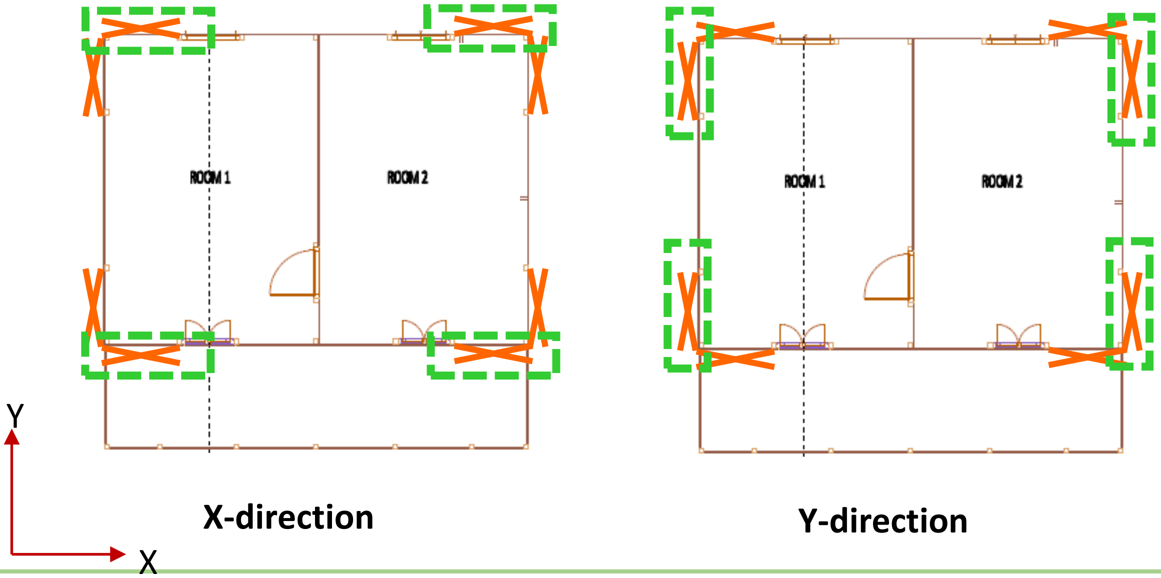
Bracing

Technical Specification

Size and Number of braces

Horizontal diagonal braces are used to resist the frame against lateral loads due to earthquake and wind.
Size and number of bracing should be consider at each X and Y direction.





X-direction **Y-direction**

Size and Number of braces

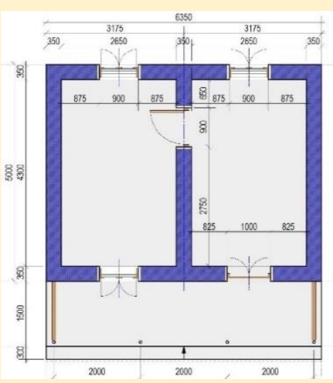
Inspection sheet base.

It shall be as per design / specification.

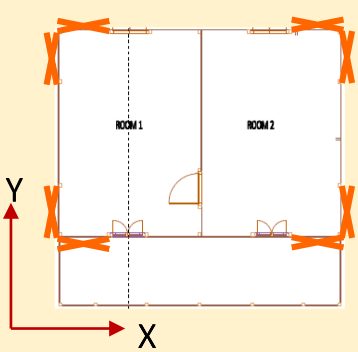
Under the following condition, simplified inspection is enabled.

- ✓ Area of building is less than 50 sqm.
- ✓ Up to 2storey without attic.
- ✓ Wall height of 1st floor is less than 2.5m
- ✓ Using light weight material of roof and wall.
- ✓ And all other requirements of each item are fulfilled.

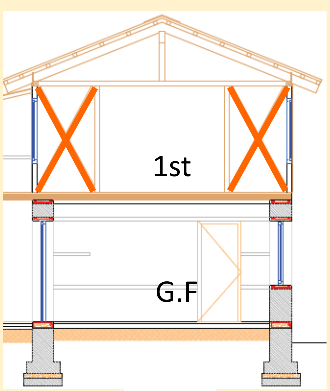
Example of size and number of brace			
Wooden Brace fixed by nail	Size	100 x 50 mm	double
	length	Minimum: 1meter	
	Number of each direction (X and Y)	4 （Located at each corner）	
Calculation	100x50: unit strength 2.6kN/m 2.6 x 2 (double) x 1 (meter) x 4 = 20.8kN		



G.F PLAN



1F PLAN

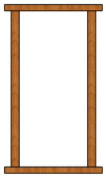
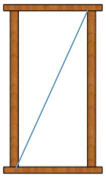
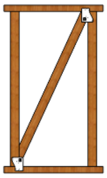
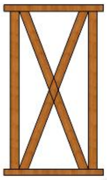


SECTION

If other brace member found to be equivalent to the strength of above example after calculation as next page.

4. Frame Action

Bracing

Method of Bracing/ wall construction				Shear Strength of Unit wall (kN/m)
	No brace			0.0
	Mud wall	Thickness less than 50mm		1.5
		Thickness 50mm~70mm		1.8
		Thickness 70mm~90mm		2.2
		Thickness more than 90mm		2.5
	Brace Steel bar Φ9			1.6
 Single brace  Double brace	Wooden Brace	90mm*15mm	Nail	1.6
		90mm*30mm	Steel Plate	2.4
			Nail	1.9
		90mm*45mm	Steel Plate	3.2
			Nail	**2.6**
		90mm*90mm	Steel Plate	4.8
	Wooden plank wall			0.8
	Structural Plywood	12mm		5.2
	Gypsum Board	9mm		1.1
	Plywood	3mm		0.9

Fundamental case: Wooden brace 100x50(double) fixed by nail:
2.6kN/m(unit strength) x 2 (double) x 1.0(meter) x 4 = 20.8kN

Option 1: Wooden brace 100 x 100(single) fixed by steel plate:
4.8kN/m(unit strength) x 1 (single) x 1.2(meter) x 4 = 23.04kN > 20.8kN OK

Option 2: Brace Steel bar Φ9 (double and 2pieces)
1.6kN/m(unit strength) x 2(pieces) x 2(double) x1.0(meter) x4 = 25.6kN > 20.8kN OK

SYMPLIFIED CALCULATION FOR SIZE AND NUMBER of BRACE

Size and Number of braces

Calculation base

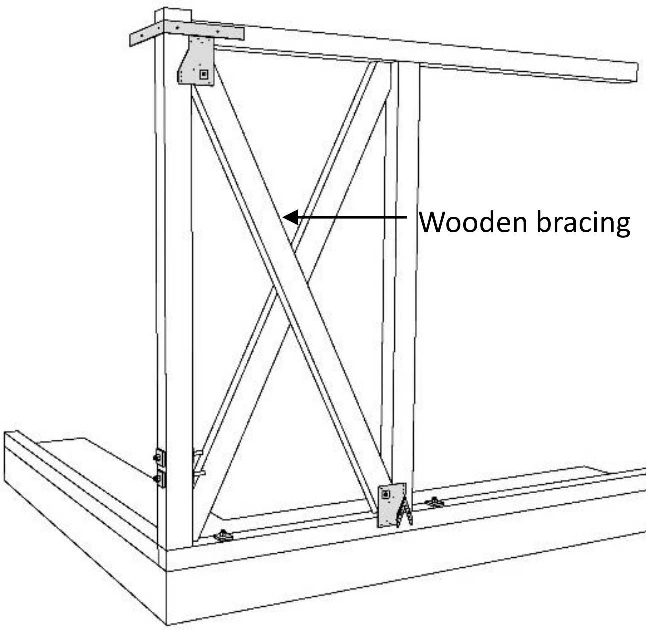
SEISMIC LOAD	Seismic coefficient	C:	Basic seismic coeffient		①	0.08	
		Z:	Zone factor		②	1	
		I:	Importance factor		③	1	
		K:	Structural performance facto		④	4	masonry structure
						2.5	frame struture
		Cd = CZIK		①x②x③x④	⑤	0.32	masonry structure
						0.2	frame struture
	Weight of building (only 1st floor)	Roof	Unit weight	Heavy	⑥	2.52	kN/sqm
				Light		0.79	kN/sqm
			Area		⑦		sqm
			Sub total	⑥x⑦	⑧		kN
		Wall	Unit weight	Heavy	⑨	2.52	kN/sqm
				Light		0.5	kN/sqm
			Volume	total length	⑩		m
				height	⑪		m
				thickness	⑫		m
			Sub total	⑨x⑩x⑪x⑫	⑬		kN
		Floor (If attic is there)	Unit weight	Heavy	⑭	2.52	kN/sqm
				Light		0.5	kN/sqm
			Area		⑮		sqm
			Sub total	⑭x⑮	⑯		kN
		TOTAL WT.		⑧+⑬+⑯	⑰		kN
	Seismic load = Cd x WT			⑤x⑰	⑱		kN

Allowable strenght	X-direction	brace	Type	refer from table	①		kN/m
				single	②	1	
				double		2	
			length		③		m
			Number		④		
			Total length	③x④	⑤		m
		Total strength		①x②x⑤	⑥		kN
	Y-direction	brace	Type	refer from table	⑦		kN/m
				single	⑧	1	
				double		2	
			length		⑨		m
			Number		⑩		
			Total length	⑨x⑩	⑪		m
		Total strength		⑦x⑧x⑪	⑫		kN

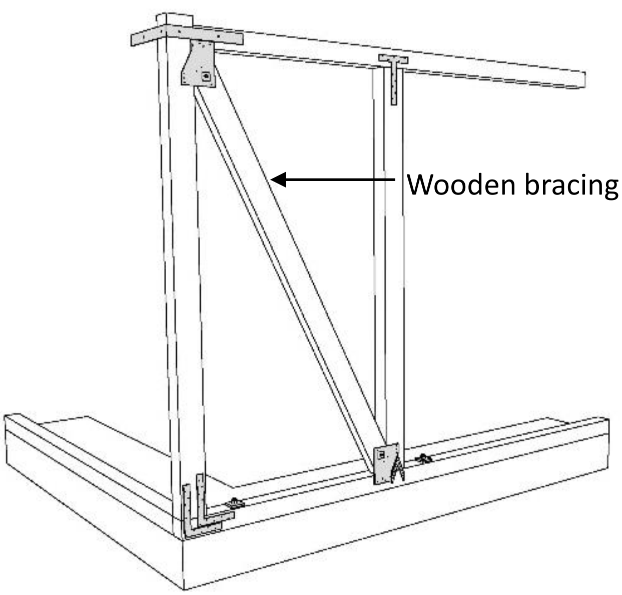
Allowable strength of each direction ⑥and ⑫ should be bigger than seismic load ⑱

Bracing

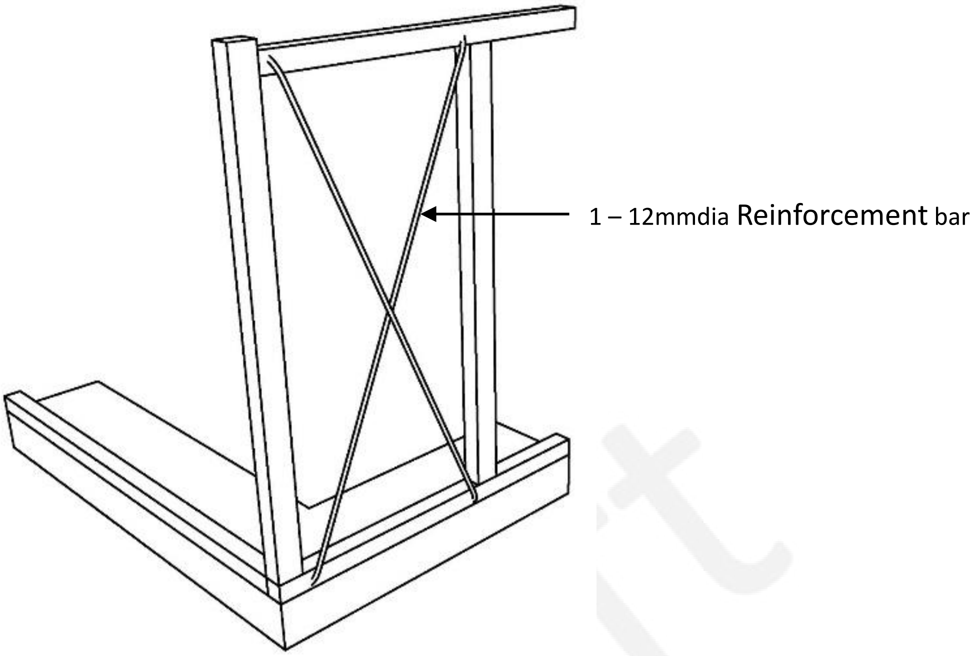
Technical Specification

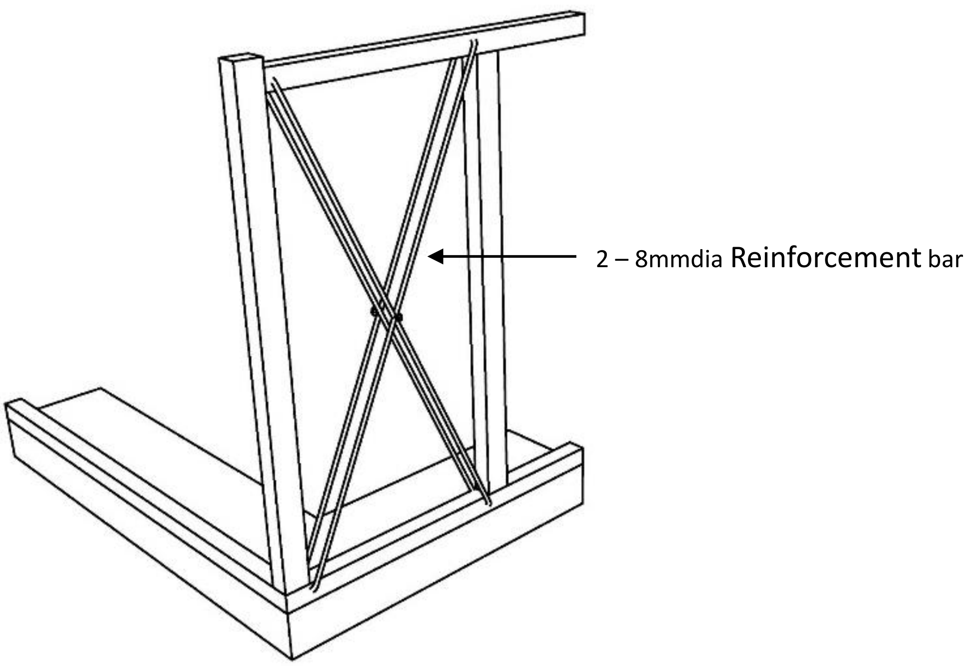


Double brace



Single brace


1 – 12mmdia Reinforcement bar


2 – 8mmdia Reinforcement bar

7. Roof

Imai 0608

Requirements

No	Category	Description		
7	Roof	Wood	Material	Use of light roof
			Connection	All member connected properly
			Bracing	For flexible diaphragm,diagonal bracing shall be considered.

Why ?

- Heavy roofs are a seismic hazard. Hence, roofs should be made as light as possible. Flexible but strong roof trusses enhance safety.
- The joints of wooden roof trusses need to be bolted together and tied with metal straps provides flexibility and prevent collapse under the forces of nature.
- In order to resist lateral forces, depending upon the structures roof structure also need cross bracing at all levels. It provides strength against lateral forces so that the building does not collapse sideways but is held together.

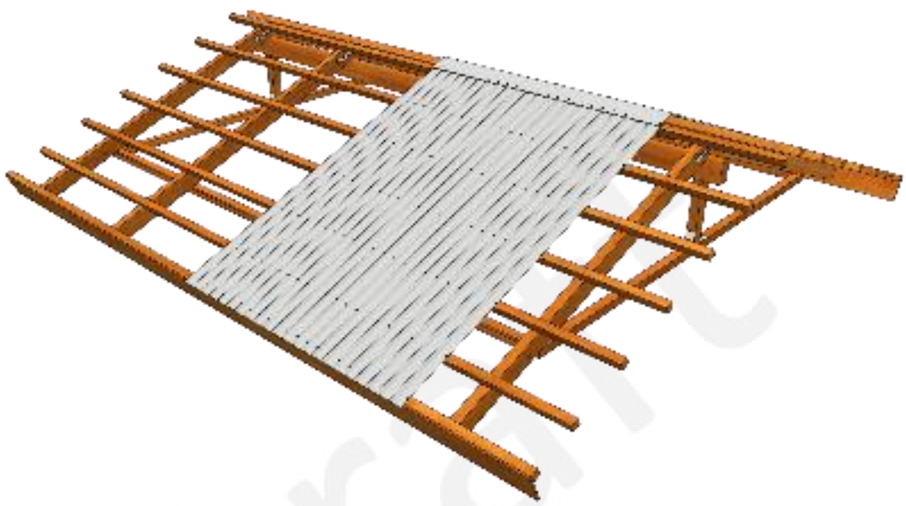
Exception

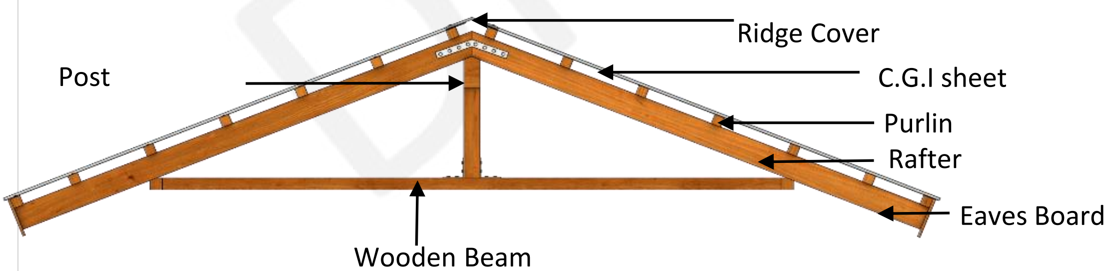
- If structure is found to be safe after structural calculation.

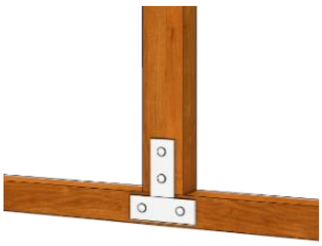
How to inspect

- The size of the main wooden member, batten can be identified by measurement.
- The spacing of the batten can also checked by the measurement whereas the connection can be checked by the observation.
- The typology of the wood can be checked by the process described in materials.
- Cross bracing in between trusses can be checked by the observation.
- Cross bracing on the roof frame can be checked by the observation.

Connection Details

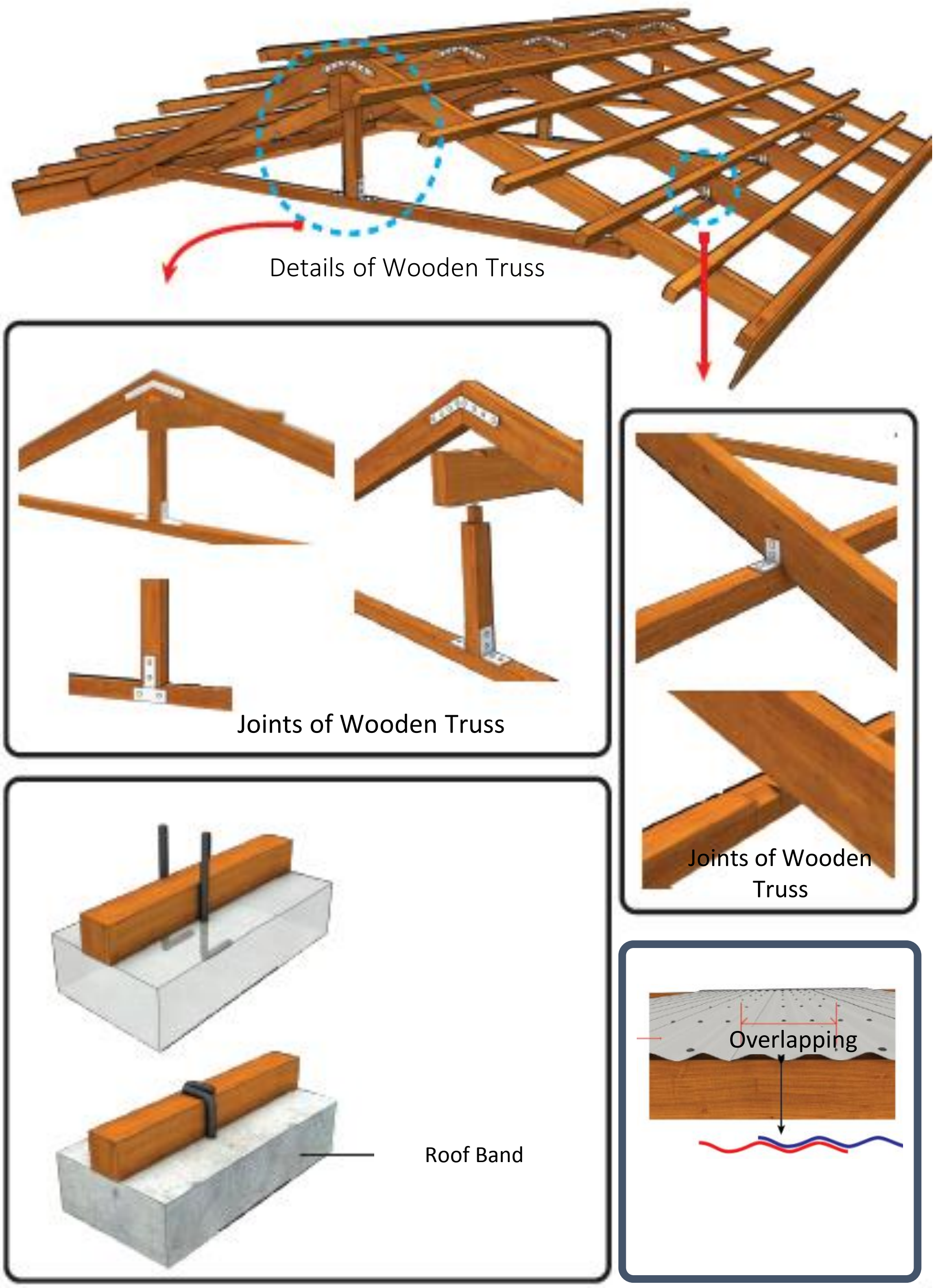






Technical Specification

Connection Details



APPENDIX

Final Inspection of Hybrid Structure

Government of Nepal
Ministry of Urban Development
Central Level Project Implementation Unit

INSPECTION SHEET OF HYBRID STRUCTURE FOR FINAL INSPECTION

Information of House Owner/Beneficiary		Date of Inspection	dd-mm-yyyy
Name:		Grand Agreement No.	
Address:	District VDC/Municipality ward tole Land plot No		

SECTION -1: DESCRIPTION PROVIDED IN THE APPLICATION TO SURVEY THE HOUSE

If use fix design from design catalogue,			Design No.	
If free design by house owner Fill construction typology from P.A form			Technique and Construction material	
			Construction of roof and materials	
Technical Assistant	☐YES, ☐NO	Organization	☐GoN, ☐NGO ( )	
Trained Masons used	☐YES, ☐NO	Soil type	☐Hard, ☐Medium, ☐Soft	

SECTION-2: DETAIL TECHNICAL INSPECTION

S.N	Category		Description			Comply YES	Comply NO	Remarks
1.	Shape of House		No. of storey	Not more than two storey		☐	☐	
			Shape of house/Span of wall	1st floor	Regular shape. The wall line of upper storey should be on the wall of lower storey. The wall line should not cantilever. Therefore, the span of wall also same as lower storey.	☐	☐	
			Height of wall	1st floor	It shall not be more than 2.5m.	☐	☐	
2.	Materials		Nail	Common wire nails shall be made of mild steel having a minimum tensile strength of 550N/mm2. Nails with appropriate diameter and length shall be provided.		☐	☐	
			Bolt	The number, diameter, length and spacing shall be as per the design/specification.		☐	☐	
			Metal plate	The number, diameter, length and spacing shall be as per the design/specification.		☐	☐	
			Rebar	High strength deformed bars with fy = 415 Mpa /500 Mpa.		☐	☐	
			Timber	Treated and well seasoned hard wood or locally available wood without knots shall be used.		☐	☐	
3.	Connection and Joints		Connection between lower and upper structure	It should be arranged properly for connecting ground floor and first floor. In SMM or BMM with timber band, the floor band and lintel band shall be properly connected.		☐	☐	
			Joints of structural member	All the structural members (Frame action) shall be properly connected by nails or bolts as per drawings/specification		☐	☐	
4.	Frame	Vertical member	It shall be properly connected with Wall plate and horizontal band.			☐	☐	
			Wood	Species	Size			
				Hard wood	(75x75)mm	☐	☐	
				Soft wood	(100X 75)mm			
				Spacing	It shall be as per design or specification	☐	☐	
			Steel ??	Equivalent to the strength of wood		☐	☐	
		Horizontal member	It shall be properly connected.			☐	☐	
			Base plate/Top plate	It shall be (100 X 75)mm.		☐	☐	
				Base plate shall be properly connected with floor band.		☐	☐	

		Brace	General	It shall be symmetrical and located at each corners.			
			Size/number	It shall be as per design/specification (*refer manual)			
5	Roof		Material	Use light roof	☐	☐	
			Connection	All member connected properly	☐	☐	
			Bracing	For flexible diaphragm, Diagonal bracing shall be considered.	☐	☐	

Others:

a) At least four number of photographs with their number
b) Tentative drawings of building:
c) After the detail description of the under constructed house, is it satisfactory to give completion certificate

Yes ☐ , No ☐

☐ It was passed through the inspection of the third inspection so the construction can be provided building completion certificate from VDC/Municipality.

☐ If was found to be corrected/retrofitted so correction order is given using Annex-6

d) Acceptation of Description provided agreeing that the technical details during inspection is correct:

House owner/Beneficiaries or representative name:..........................Signature:.........................

Relationship with house owner (In case of representative):......................Date:...............................

e) Submit for Approval of the technical inspection:...

MOUD-DLPIU Supervisor:...

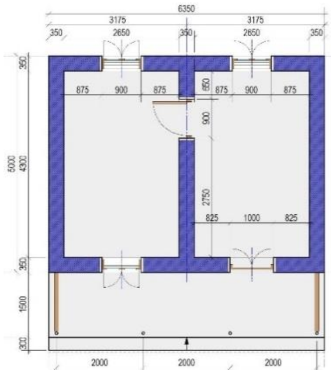
Name.. Designation:..

Signature.. Date..

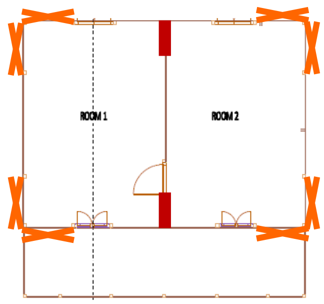
f) Approved by:

MOUD DLPIU Supervision Engineer.. Designation:..

Signature.. Date..

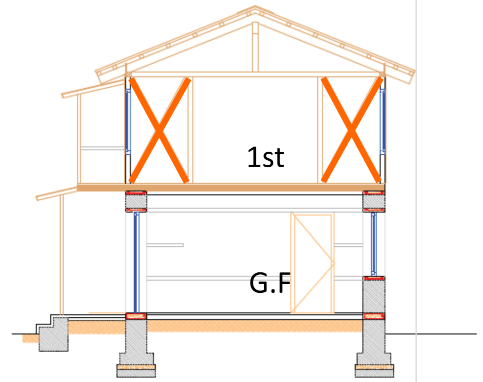


G.F PLAN

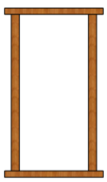
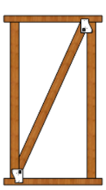
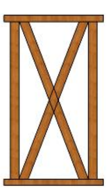
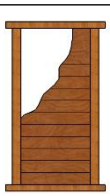


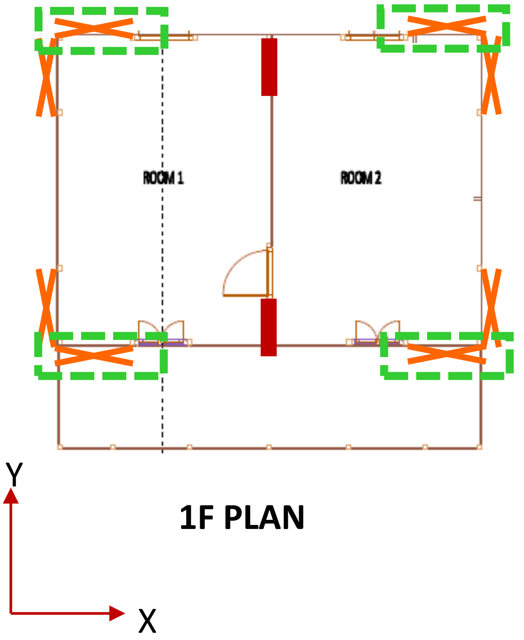
1F PLAN

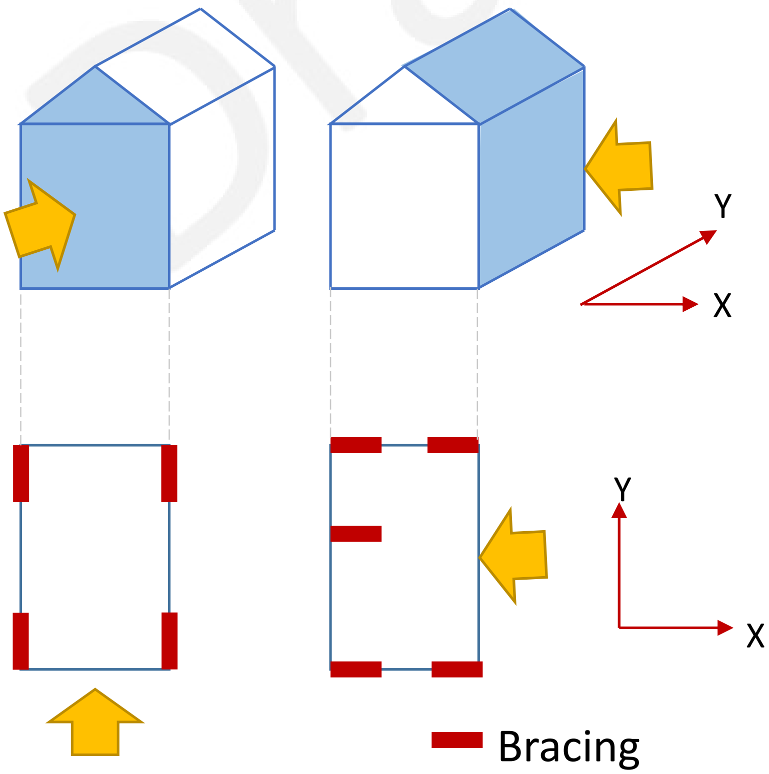
Brace: 100mm x 50mm double



SECTION

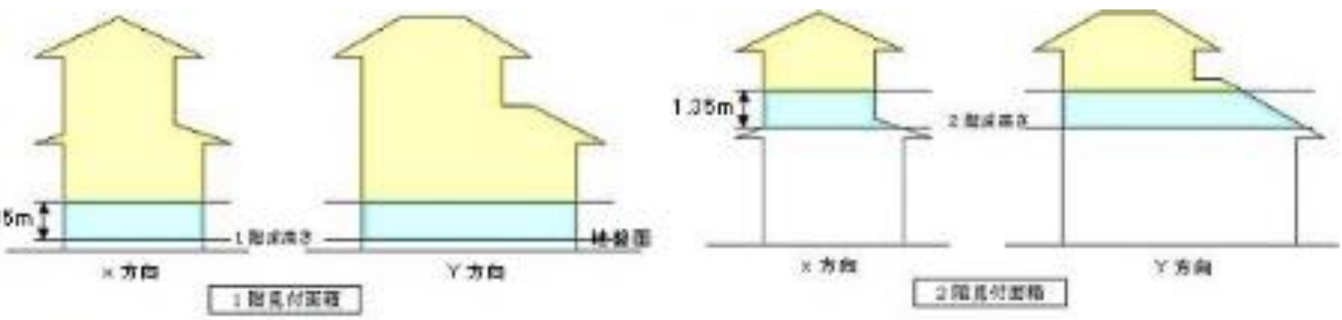
	Method of Bracing/ wall construction			Shear Strength of Unit wall (kN/m)
	No brace			0.0
	Mud wall	Thickness less than 50mm		1.5
		Thickness 50mm~70mm		1.8
		Thickness 70mm~90mm		2.2
		Thickness more than 90mm		2.5
	Brace Steel bar Φ9			1.6
 Single brace  Double brace	Wooden Brace	90mm*15mm	Nail	1.6
		90mm*30mm	Steel Plate	2.4
			Nail	1.9
		90mm*45mm	Steel Plate	3.2
			Nail	2.6
		90mm*90mm	Steel Plate	4.8
	Wooden plank wall			0.8
	Structural Plywood	12mm		5.2
	Gypsum Board	9mm		1.1
	Plywood	3mm		0.9


ROOM 1
ROOM 2
1F PLAN
Y
X


Y
X
Y
X
Bracing

STEP2. Wind load

S code: Maximum wind speed: 55m/3s
apanese Building code: Average wind speed 30~35m/10min



❑ SEISMIC LOAD

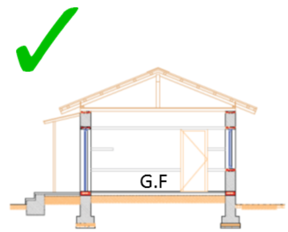
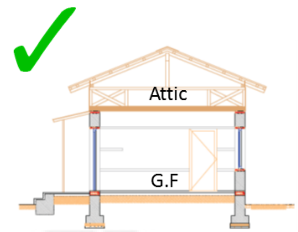
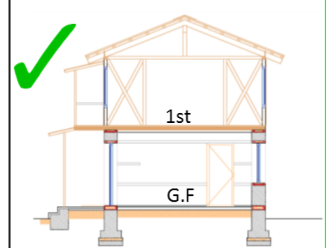
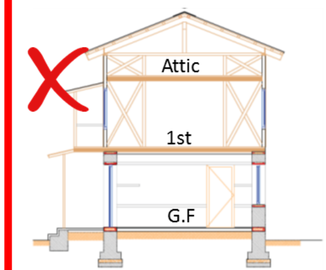
The resultant lateral force or seismic load is represented by the force F in fig.2.6. It is distinctly different from dead, live, snow, wind and impact loads. The horizontal ground motion action is similar to the effect of a horizontal force acting on the building, hence the term “Seismic Load” or “Lateral Load” is used. As the base of the building moves in an extremely complicated manner, inertia forces are created throughout the mass of the building and its contents. It is these reversible forces that cause the building to move and sustain damage or collapse.

An additional and uplift vertical load effect is caused on slabs, beams cantilevers and columns due to vertical vibrations, which may cause damage. Being reversible, at certain instants of time the effective load is increased, at others it is decrease.

Earthquake loads are dynamic and impossible to predict precisely in advance since every earthquake exhibits different characteristics. The following equivalent lateral force *F* that is used for seismic design, is expressed as the product o the mass of the structure *m* and the acceleration *a* or the seismic coefficient *k* and the weight of the structure *W:*
$F = ma = kW$
W is the total weight of the supper-structure of a building including its contents. The inertia forces are proportional to the mass (or weight) of the building and only building elements or contents that possess mass will give rise to seismic force on the building. Therefore, the lighter the material, the smaller will be the seismic force.

Failure of ground floor

Failure of 1st floor

STEP 1: CHECKING SEISMIC LAOD			ONE STOREY	ONE STOREY+ATTIC	TWO STOREY	TWO STOREY + ATTIC
HYBRID STRUCURE (JICA MODEL) Floor Area: 6.35m*5.0=31.75m²			 G.F:SMM+Wooden band	 G.F:SMM+Wooden band Attic:Timber frame	 G.F:SMM+Wooden band 1F:Timber frame	 G.F:SMM+Wooden band 1F:Timber frame
Area of wall on each direction	Attic	X-direction		L=12.7m A=1.27 m²		L=12.7m A=1.27 m²
		Y-direction		L=10 A=1m²		L=10 A=1m²
	1st	X-direction			L= 12.7 m A=1.27 m²	L= 12.7 m A=1.27 m²
		Y-direction			L=15m A= 1.5 m²	L=15m A= 1.5 m²
	G.F	X-direction	L= 8.4m, A=2.95m²	L= 8.4m, A=2.95m²	L= 8.4m, A=2.95m²	L= 8.4m, A=2.95m²
		Y-direction	L=13.05m, A=4.57m²	L=13.05m, A=4.57m²	L=13.05m, A=4.57m²	L=13.05m, A=4.57m²
Combination Load	Roof		0.79 KN/m² X 6.8m X 8.15m= **43.80 kN**	0.79 KN/m² X 6.8m X 8.15m= **43.80 kN**	0.79 KN/m² X 6.8m X 8.15m= **43.80 kN**	0.79 KN/m² X 6.8m X 8.15m= **43.80 kN**
	Floor	Attic		2.77 KN/m² X 6.0m X 4.65m= **77.28 kN**		2.77 KN/m² X 6.0m X 4.65m= **77.28 kN**
		1st			2.52 KN/m² X 6.0m X 4.65m= **69.75 kN**	2.52 KN/m² X 6.0m X 4.65m= **69.75 kN**
	Wall	Attic		0.50 KN/m² X 26.7m = **13.35 kN**		0.50 KN/m² X 26.7m = **13.35 kN**
		1st			0.50 KN/m² X 69.09m² = **34.54 KN**	0.50 KN/m² X 65.09m = **32.54 kN**
		G.F	9.35 KN/m² X {(6.0m X 2+ 4.65 X 3)X2.7 –(0.9m X 1.35)X3- (0.9mX2.1m)X2}= **585.07 kN**	9.35 KN/m² X {(6.0m X 2+ 4.65 X 3)X2.7 –(0.9m X 1.35)X3- (0.9mX2.1m)X2}= **585.07 kN**	9.35 KN/m² X {(6.0m X 2+ 4.65 X 3)X2.7 –(0.9m X 1.35)X3- (0.9mX2.1m)X2}= **585.07 kN**	9.35 KN/m² X {(6.0m X 2+ 4.65 X 3)X2.7 –(0.9m X 1.35)X3- (0.9mX2.1m)X2}= **585.07 kN**
	TOTAL WT. (WT of 1st)		**628.87 kN**	**719.50 kN (57.15kN)**	**733.16 kN (78.34kN)**	**821.80 kN (166.97kN)**
SEISMIC LOAD		G.F	Cd=0.08*1*1*4=0.32, k=4(masonry) **V=628.87kN*0.32=201.24 kN**	Cd=0.08*1*1*4=0.32, k=4(masonry) **V=719.50kN*0.32=230.24 kN**	V=Cd X Wt, Cd=0.08*1*1*4=0.32 **V=733.16*0.32=234.61 kN**	V=Cd X Wt, Cd=0.08*1*1*4=0.32 **V=821.80kN*0.32=262.98 kN**
		1st		Cd=0.08*1*1*2.5=0.2, k=2.5(Frame) **V=57.15*0.2=11.43 kN**	Cd=0.08*1*1*2.5=0.2 **V=78.34*0.2=15.67 kN**	Cd=0.08*1*1*2.5=0.2 **V=166.97*0.2=33.39 kN**
Allowable strength	G.F	X-direction	(2950000*0.096)/1000 =283.2 KN > 201.24kN ⇒OK	(2950000*0.096)/1000 =283.2 KN > 230.24 kN ⇒OK	(2950000*0.096)/1000 =283.2 KN > 234.61 kN ⇒OK	(2950000*0.096)/1000 =283.2 KN > 262.98 kN ⇒OK
		Y-direction	(4570000*0.096)/1000 =438.72 KN > 201.24 kN ⇒OK	(4570000*0.096)/1000 =438.72 KN > 230.24 kN ⇒OK	(4570000*0.096)/1000 =438.72 KN > 234.61 kN ⇒OK	(4570000*0.096)/1000 =438.72 KN > 262.98 kN ⇒OK
	1st/ Attic	X-direction		Brace (X:100*50) =2.6*2*4 =20.8 kN > 11.43 kN ⇒OK	Brace (X:100*50) =2.6*2*4 =20.8 kN > 15.67 kN ⇒OK	Brace (X:100*50) =2.6*2*4 =20.8 kN < 33.39 kN ⇒NG
		Y-direction		Brace (X:100*50) =2.6*2*4 =20.8 kN > 11.43 kN ⇒OK	Brace (X:100*50) =2.6*2*4 =20.8 kN > 15.67 kN ⇒OK	Brace (X:100*50) =2.6*2*4 =20.8 kN < 33.39 kN ⇒NG

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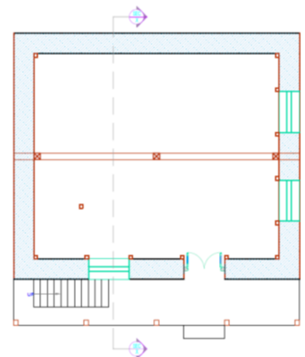
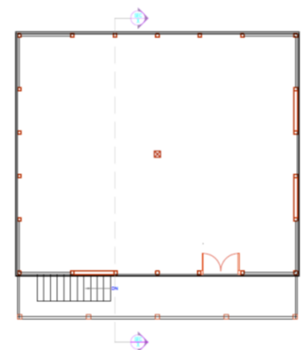
STEP 1: CHECKING SEISMIC LAOD

HYBRID STRUCURE (NSET MODEL)			**TWO STOREY**	
			 **G.F PLAN**  **1F PLAN**  **SECTION A-A'**	
			GL:SMM+Wooden band, 1F:Timber frame Floor Area: 6.3m*5.4=34.02m^{2,} Wall thickness: 0.45m, Height: GL=3.0m, 1st= 2.5m	
Area of wall on each direction	**1st**	**X-direction**	L= 12.6 m A=1.276m^2	
		Y-direction	L=10.8m A= 1.08 m^2	
	G.F	**X-direction**	L= 8.73m, A=3.92m^2	
		Y-direction	L=7.74m, A=3.48m^2	
Combination Load	**Roof**		0.79 KN/m^2 X 8.4m X 7.5m= **49.77 kN**	
	Floor	**1st**	2.52 KN/m^2 X 6.3m X 5.4m= **86.07 kN**	
	Wall	**1st**	0.50 KN/m^2 X 63.9m^2 = **31.95 KN**	
		G.F	9.35 KN/m^2 X {(5.85m X 2+ 4.95 X 2)X3.0 –(0.9m X 1.35)X3- (0.9mX2.1m)}= **585.51 kN**	
	TOTAL WT. (WT of 1st)		**753.30 kN (81.72 kN)**	
Design Method			**Working Stress Method**	**Limit State Method**
SEISMIC LOAD		**G.F**	V=Cd*Wt, Cd=0.08*1*1*4=0.32 **V=753.30*0.32=241.06 kN**	V=Cd*Wt *1.25, Cd=0.08*1*1*4=0.32 **V=941.63 kN*0.32=301.32 kN**
		1st	V=Cd*Wt, Cd=0.08*1*1*2.5=0.2 **V=81.72*0.2=16.34 kN**	V=Cd*Wt *1.25, Cd=0.08*1*1*2.5=0.2 **V=102.15*0.2=20.43 kN**
Shear Strength	**G.F**	**X-direction**	(3920000*0.096)/1000 =376.32 KN > 241.06 kN ⇒OK	(3920000*0.096*1.5)/1000 =564.48KN > 301.32 kN ⇒OK
		Y-direction	(3480000*0.096)/1000 =334.08 KN > 241.06 kN ⇒OK	(3480000*0.096*1.5)/1000 =501.12KN > 301.32 kN ⇒OK
	1st	**X-direction**	Brace (X:100*50) =2.6*2*4 =20.8 kN > 16.34 kN ⇒OK	Brace (X:100*50) =2.6*2*4*1.5 =31.2 kN > 20.43 kN ⇒OK
		Y-direction	Brace (X:100*50) =2.6*2*4 =20.8 kN > 16.34 kN ⇒OK	Brace (X:100*50) =2.6*2*4*1.5 =31.2 kN > 20.43 kN ⇒OK

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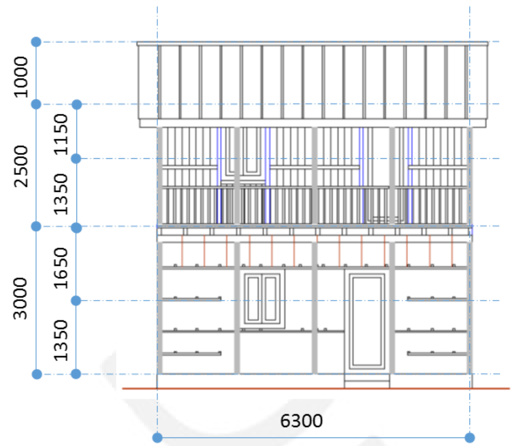
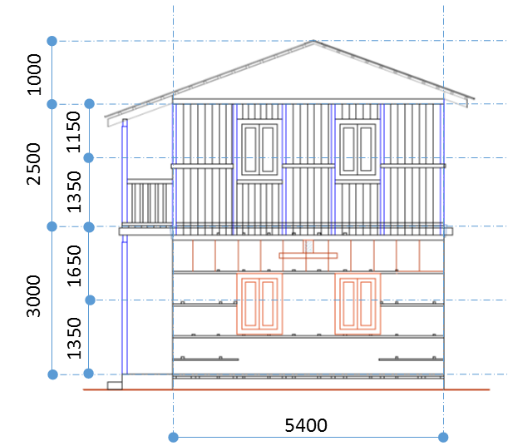
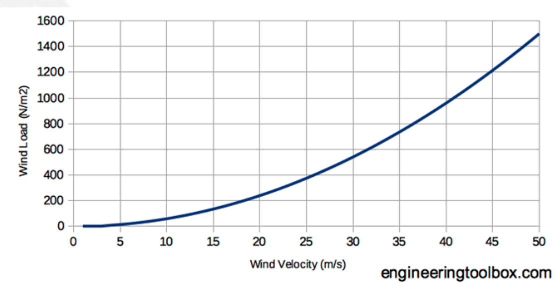
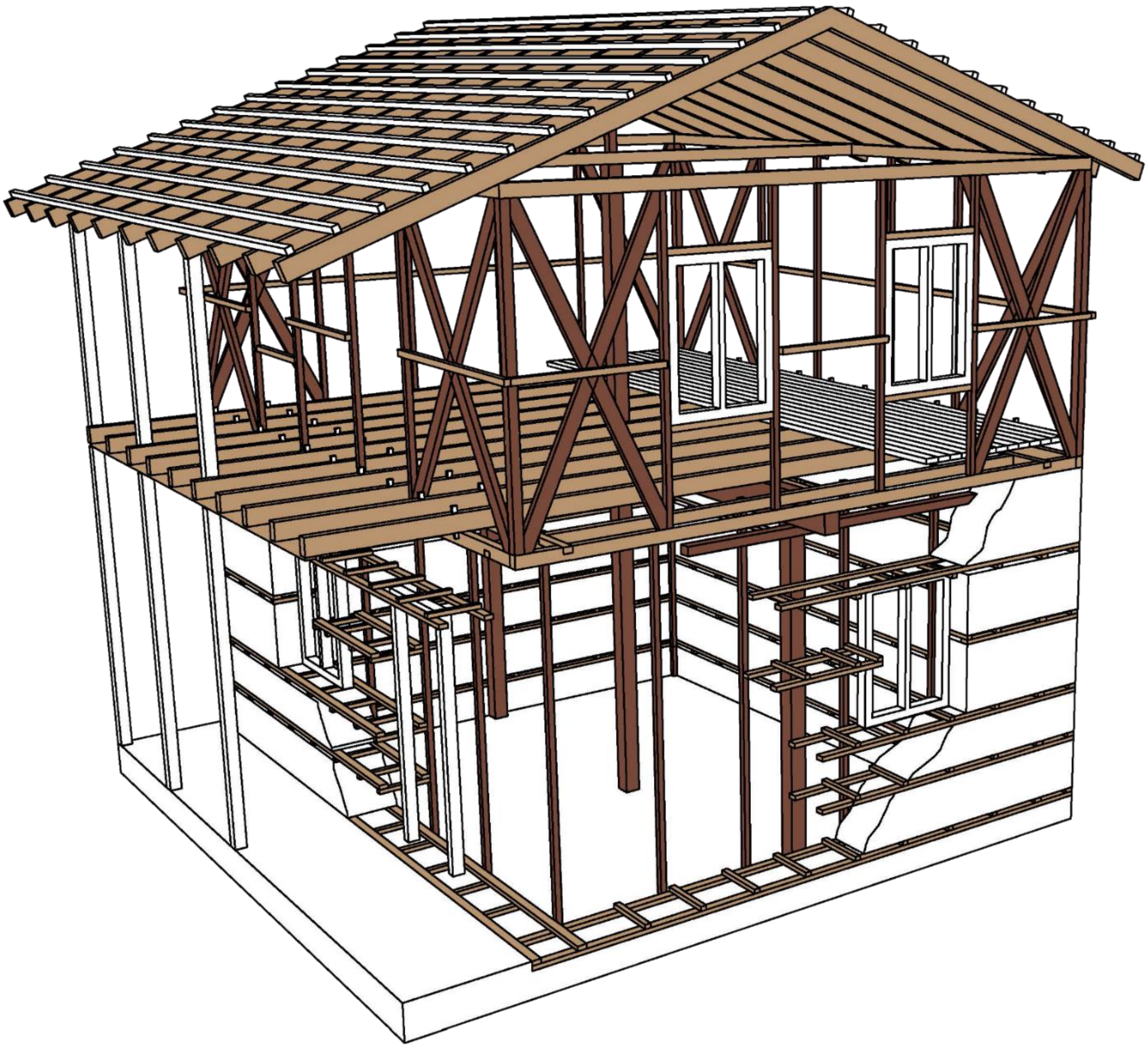
APPENDIX: Wall ratio calculation

STEP 1: CHECKING WIND LAOD

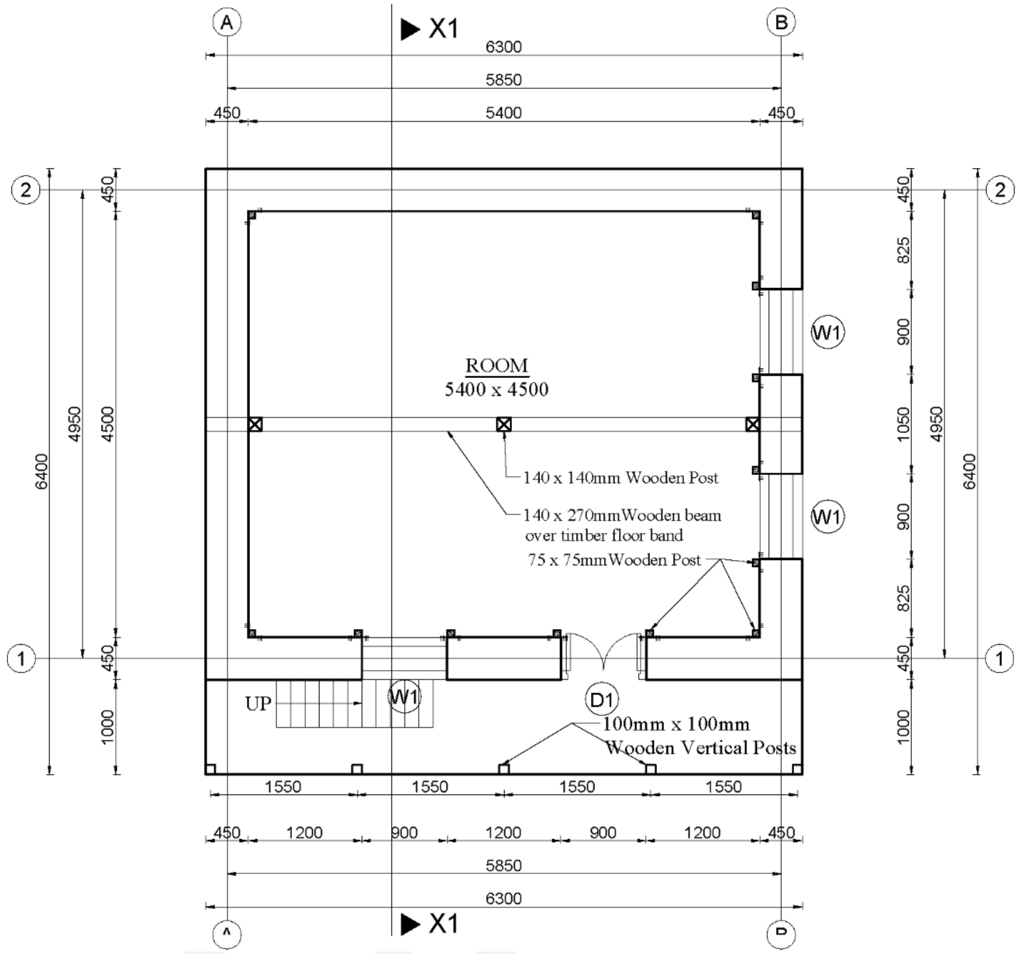
			X-direction	Y-direction
Elevation of model				
Related wall Area	1st floor		7.5m*1.0m+6.3m*1.15m = 14.75m²	5.4m*1.0m/2+5.4m*1.15m = 8.91m²
	G.R floor		7.5m*1.0m +6.3m*(2.5m+1.65m) = 33.65m²	5.4m*1.0m/2+5.4m*(2.5m+1.65m) = 25.11m²
Wall ratio	1st floor	Requirement	14.75m² *0.98kN/m=14.46kN	8.91m² *0.98kN/m=8.73kN
		Capacity	Brace (X:100*50) =2.6*2*4=20.8 kN	Brace (X:100*50) =2.6*2*4=20.8 kN
		Result	20.8 kN > 14.64kN ⇒OK	20.8 kN > 8.73kN ⇒OK
	G.R floor	Requirement	33.65m² *0.98kN/m=35.92kN	25.11m² *0.98kN/m=25.60kN
		Capacity	(3920000*0.096)/1000=376.32 KN	(3920000*0.096*1.5)/1000=564.48KN
		Result	376.32 KN > 35.92 kN ⇒OK	564.48KN >25.60k ⇒OK
Notes			1. **The requirement of wall ratio, In case of Japan, The requirement of the wall length is 50cm/m2 based on 1.96kN/m. Therefore, The requirement of wall ratio will be 0.98kN as per related wall area. The requirement is 0.98KN/m2.** 2. **Estimated wind speed : In Japan . Estimate wind speed is 34m/s (10min average), and In Indian Standard 55m/s (3second average). It can assume, 3 second average is equivalent 1.5 ~ 2.0 times of 10 min average. Therefore, Japanese method also applicable for wind check for ordinary house.**	 **Relationship between wind velocity and wind load** 14

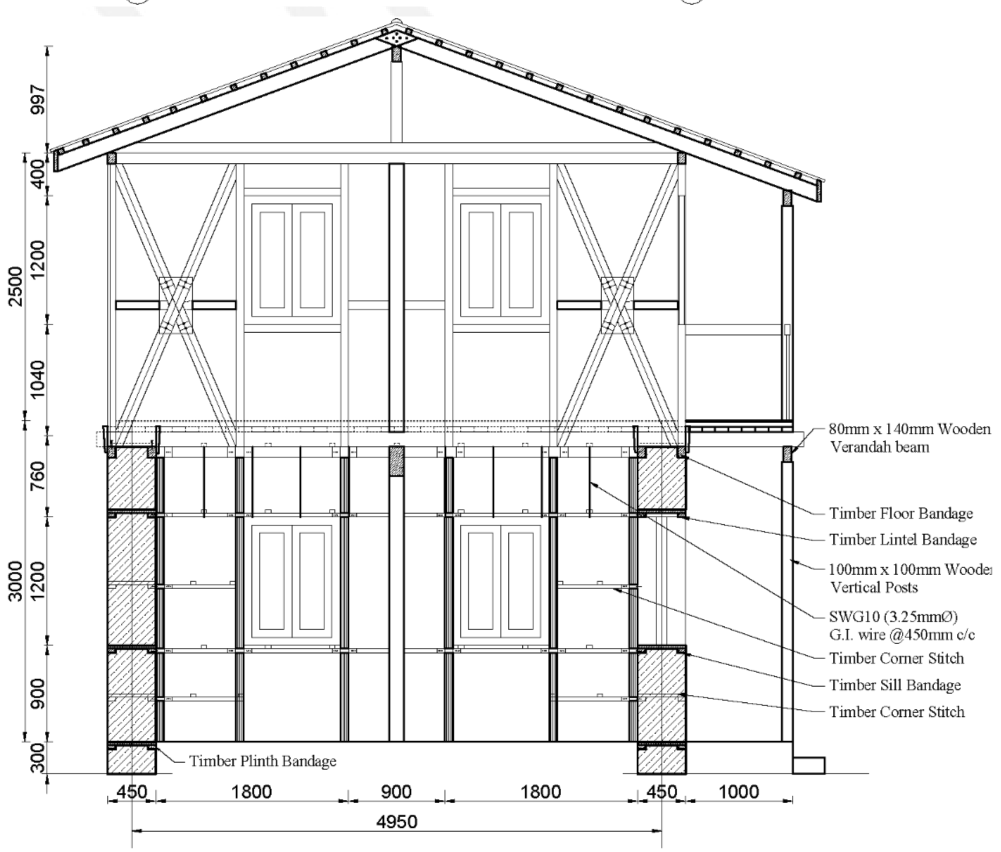


Building description:

Sample building used in this calculation is a Two Story Mix Structural System Building with Low Strength Stone Masonry in Ground Floor and Timber Structure in Upper Floor.

Building type: Residential building
Plan shape: Rectangular
Plinth area: 34.02 sq.m.
Number of storey: Two (Ground floor low strength masonry and upper storey timber structure)
Total height: 5.5m from plinth level
Inter storey height: Ground floor 3m and upper floor 2.5m
Building system: Mixed: Ground floor low strength load bearing masonry and upper floor timber structure.


ROOM
5400 x 4500
140 x 140mm Wooden Post
140 x 270mm Wooden beam over timber floor band
75 x 75mm Wooden Post
100mm x 100mm Wooden Vertical Posts
UP
W1
D1
X1
6300
5850
5400
450
1550
1200
900
4950
4500
6400
825
1050
1000


80mm x 140mm Wooden Verandah beam
Timber Floor Bandage
Timber Lintel Bandage
100mm x 100mm Wooden Vertical Posts
SWG10 (3.25mmØ) G.I. wire @450mm c/c
Timber Corner Stitch
Timber Sill Bandage
Timber Corner Stitch
Timber Plinth Bandage
997
400
2500
1200
1040
760
3000
900
300
450
1800
1000
4950

Basic Assumption

1.Unit weight of Materials

Unit weight of the materials is taken from IS 875 part I

Unit Weight of Masonry=	19.00	KN/m3
Unit Weight of Timber=	5.75	KN/m3
Unit Weight of Floor Covering =	19.00	KN/m3
Weight per m2 of CGI Sheet =	0.13	KN/m2

i. Material Properties

a. Stone Masonry Wall

Modulus of Elasticity (E) = 74 Mpa

Allowable Compressive Strength = 0.47 Mpa

(Calculated From IS 1905: Code of Practice for structural use of unreinforced masonry)

Allowable Tensile Strength= Neglected

Allowable Shear Strength= 0.096Mpa (From Shake table test of China)

b. Timber: Soft Wood (Chir) (As per NBC 112:1994)

Allowable Tensile Strength (ft)	=	6.9 Mpa
Allowable Compressive Strength (fc)	=	5.5 Mpa
Allowable Shear Strength (fs)	=	0.6 Mpa

Load calculation

Dead Load and Live Load:

Dead Load

Unit Weight of Masonry=	19.00	KN/m3
Unit Wt. of RCC=	25.00	KN/m3
Unit Weight Timber=	5.75	KN/m3
Unit Weight of Floor Covering =	16.00	KN/m3
Weight per m2 of CGI Sheet =	0.13	KN/m2
Load from Floor		
Length of Building	6.30	m
Width of Building	5.40	m
Width of Timber =	0.08	m
Depth of Timber =	0.10	m
Spacing of Timber in X direction	0.59	m
Total Length of timber(m)	58.15	m
Total Weight of timber =	2.51	KN
Depth of Floor Covering =	0.15	m
Weight per area of Floor Covering =	16.00	KN/m2
Total Dead Weight of Floor =	84.16	KN
Weight Density per m2 of Floor =	2.47	KN/m2

Load from Roof

Length of Building	6.30	m
Width of Building	5.40	m
Height of Roof =	1.15	m
Inclined Length of One roof =	3.99	m
Spacing of Purlin=	1.00	m
Width of purlin =	0.05	m
Depth of Purlin =	0.05	m
Length of purlin =	34.02	m
Spacing of Rafter =	1.00	m
Width of Rafter =	0.080	m
Depth of Rafter =	0.14	m
Length of Rafter =	50.24	m
Weight of Purlin =	0.49	KN
Weight of Rafter =	3.24	KN
Weight of CGI Sheet =	7.51	KN
Total Wt of Roof =	11.23	KN
Wt per M2 of Roof =	0.33	KN/m2

Live Load

Live load Intensity on floor =	3.00	KN/m2
Live load Intensity on Roof =	0.49	KN/m2

Load on Floor Rafter

Live load =	1.755	KN/m
Dead Load =	1.447	KN/m

Load on roof rafter

Live load =	0.287	KN/m
Dead Load =	0.193	KN/m

Timber Plank Wall load

Width of plank =	0.038	m
Height =	2.5	m
Unit weight =	16	KN/m3
Weight =	1.52	KN/m

Seismic Load Calculation:

Seismic load calculation is done as per NBC: 105.

Table : Seismic Load Calculation

Zone Factor	Z	1		
Importance factor	I	1		cl 8.1.7, table 8.1, other structures
Structural performance factor	K	2.5		cl 8.1.8, table 8.2, Reinforced Masonry Building (Taken average of Unreinforced and RCC framed Structure)
Height of the building	h	5.5	m	
Dimension of the building along X	D_x	6.300	m	
Dimension of the building along Y	D_y	5.4	m	
Time period of the building along X	$T_x = 0.09h/\sqrt{D_x}$	0.197	sec	cl 7.3
Time period of the building along Y	$T_y = 0.09h/\sqrt{D_y}$	0.213	sec	cl 7.3
Soil type	Soft type	Type III		cl 8.1.5
Basic seismic coefficient along X	C_x	0.08		cl 8.1.4, fig 8.1
Basic seismic coefficient along Y	C_y	0.08		cl 8.1.4, fig 8.2
Design horizontal seismic coefficient	C_d = CZIK	0.2		cl 8.1.1

Wind Load Calculation:

Wind load is calculated as per NBC 104:1994 Wind load and IS 875(Part 3):1987.

Design velocity of Wind (Vb) = 55 m/s

(Considering Upper Part of Nepal, Figure 1.1, NBC 101:1994)

Probability factor (K1) = 1

(For general building and structure with wind speed =55 m/s, Table 1, Clause 5.3.1, IS 875(Part3):1987)

Terrine Category = 1 (Taking most severe)

Building Class = A (Lateral Dimension less than 20m)

For Building Class A and terrine category 1, Height is smaller than 10m)

Terrine, height and structure size factor (K2) =1.05

Assuming Slope angle = 20^0

C= 0.36 (Slope angle = $20^0 > 17^{0)}$

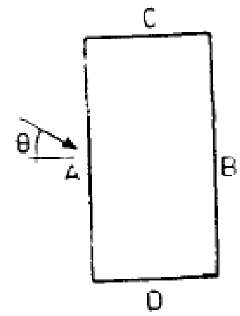
S= 1 (Most Severe Case)

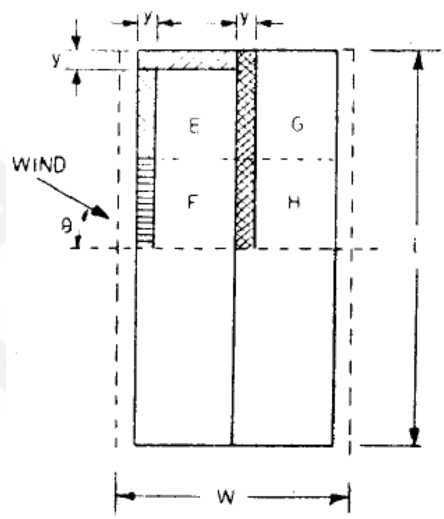
Topography Factor (K3) = 1+CS =1+0.36*1 =1.36

Base on the building Dimension following coefficient is calculated

Load on Wall/Cladding of the building

Wall	Angle of wind $=0^0$			Angle of Wind $=90^0$		
	Cpe	Cpi	Cp	Cpe	Cpi	Cp
A	0.7	0.5	1.2	-0.6	-0.5	-1.1
B	-0.25	-0.5	-0.75	-0.6	-0.5	-1.1
C	-0.6	-0.5	-1.1	0.7	0.5	1.2
D	-0.6	-0.5	-1.1	-0.25	-0.5	-0.75





Load on Pitched Roof (Roof angle = 23^0)

Portion	Angle of wind $=0^0$			Angle of Wind $=90^0$		
	Cpe	Cpi	Cp	Cpe	Cpi	Cp
Wind Ward	-0.55	-0.5	-1.05	-0.8	-0.5	-1.3
Leeward	-0.5	-0.5	-1	-0.65	-0.5	-1.15

DETAILED STRUCTURAL ANALYSIS

•Finite Element Modeling of the building of Supreme is done by using the structural analysis and design software program SAP 2000vs19. For the analysis of the system, whole building is modelled. Load bearing stone masonry walls modelled as single layered shell elements. Timber member of roof and floor is modeled as the line element with 4 degree of freedom in each node ie. Pinned joint.

Nepal National Building Code NBC 105:1994 is used for the seismic load calculations and IS 875(part 3):1987 and NBC 104:1994 is used for the wind load calculation. 3D view of the analytical model is shown in Figure.

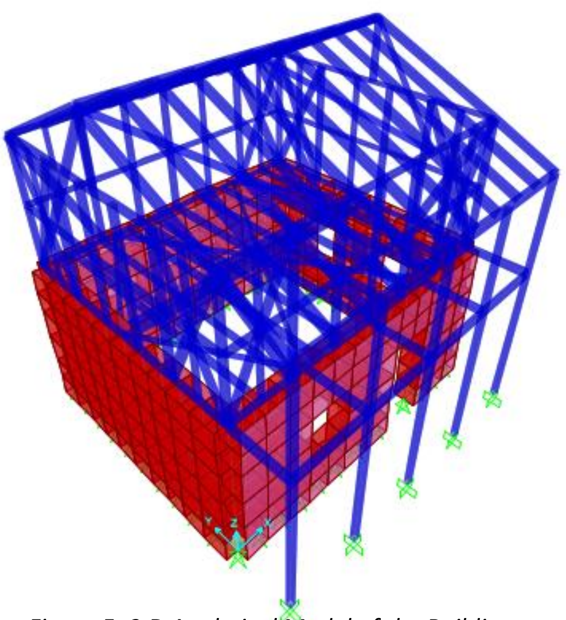


Figure 5: 3 D Analytical Model of the Building

SEISMIC ANALYSIS

The seismic analysis is a part of the detailed evaluation of an existing building. The steps involve in developing a computational model of the building include applying the external forces, calculating the internal forces in the members of the building, identifying deformations and capacity of the members and building, and finally interpreting the results. The structural analysis is carried out with the help of the available drawings and SAP 2000 vs 19. Seismic coefficient method is used to analyze the building.

Calculation of Base Shear

Base Shear in the building

Load Pattern	Type	Direction	C	Weight Used (KN)	Base Shear (KN)
EQX	Seismic	X	0.2	612.789	122.558
EQY	Seismic	Y	0.2	612.789	122.558

Load Combination for analysis of the building

Load Considered for the analysis:

- Dead Load (DL)
- Live Load (LL)
- Earthquake Load (EQ)

Load Combination: For Working Stress method As per NBC 105:

- DL+LL
- 0.7DL+EQx
- 0.7DL-EQx
- 0.7DL+EQy
- 0.7DL-EQy
- DL+LL+EQx
- DL+LL-EQx
- DL+LL+EQy
- DL+LL-Eqy

Check for Deflection

Deflection check is done as per the requirement of NBC 105 Clause 9. The design lateral deformations is taken as the deformations resulting from the application of the design force, multiplied by the factor 5/K. As per NBC 105 Clause 9.3 .The ratio of the inter-story deflection to the corresponding story height shall not exceed 0.010 nor shall the inter-story deflection exceed 60 mm. Detail check pf deflection is shown in table below.

Check For deflection for Load Case EQX

Story	Maximum Deflection	Story height	Structural Performance Factor	Design Lateral Deflection	Interstory Drift	Interstory Drift Ratio	Permissible interstory Drift Ratio	Status
	(D)mm	(h) mm	(K)	(D*5/K) mm	mm			
1	3.85	3000	2.5	7.7	7.7	0.0025	0.01	OK
2	7.19	2500	2.5	14.38	6.68	0.0027	0.01	OK

Check For deflection for Load Case EQY

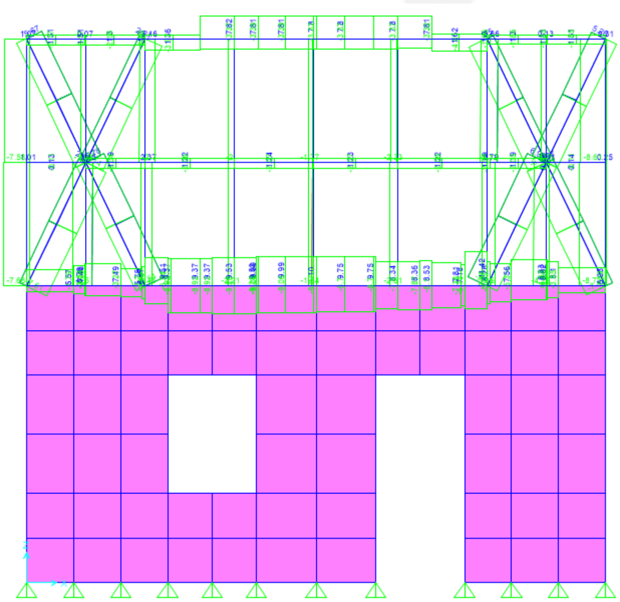
Story	Maximum Deflection	Story height	Structural Performance Factor	Design Lateral Deflection	Interstory Drift	Interstory Drift Ratio	Permissible interstory Drift Ratio	Status
	(D)mm	(h) mm	(K)	(D*5/K) mm	mm			
1	4.59	3000	2.5	9.18	9.18	0.0031	0.01	OK
2	8.2	2500	2.5	16.4	7.22	0.0029	0.01	OK

Modeling output for existing building

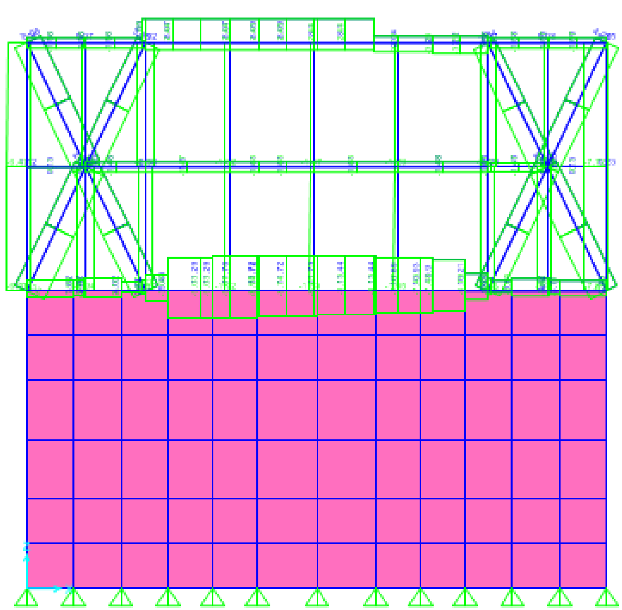
(Mention that the masonry building has been checked for stresses and found to be safe. Details of analysis and design is shown only for the upper story timber structure)

Initially, building is modeled and axial forces for different load combination is studied. The axial force develop for envelope combination is shown below.

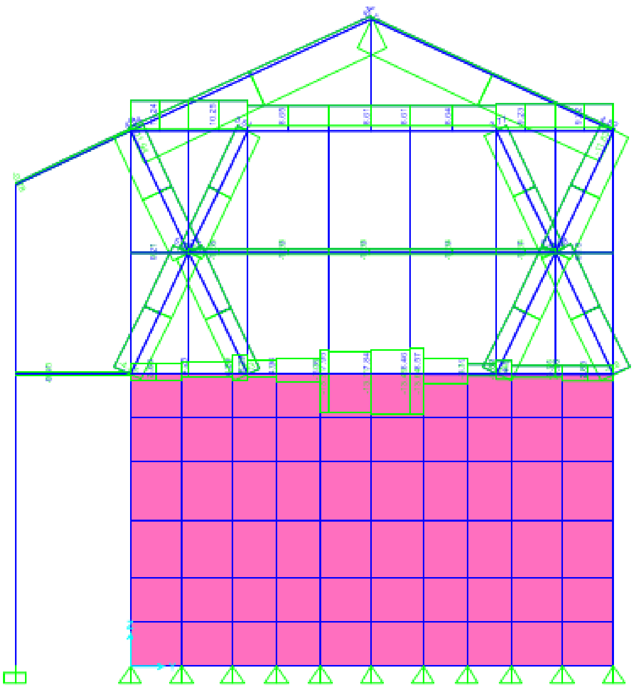
Axial Forces



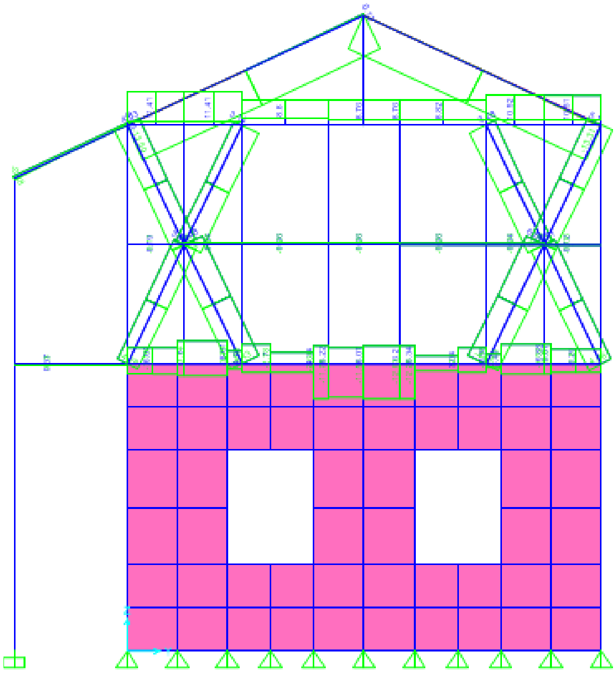
Envelope Axial Force (X-direction: Grid 1-1)



Envelope Axial Force (X-direction: Grid 2-2)



Envelope Axial Force (Y-direction: Grid A-A)



Envelope Axial Force (Y-direction-B-B)

WIND LOAD ANALYSIS

Wind load analysis is done as per IS 875(part 3) :1987 and NBC 104 :1994 .The steps involve in developing a computational model of the building include applying the external forces, calculating the internal forces in the members of the building, identifying deformations and capacity of the members and building, and finally interpreting the results. The structural analysis is carried out with the help of the available drawings and SAP 2000 vs 19.

1.Calculation of Wind load

2.Joint Reaction at the base of timber floor.

3.Load Combination for analysis of the building

4.Load Considered for the analysis:

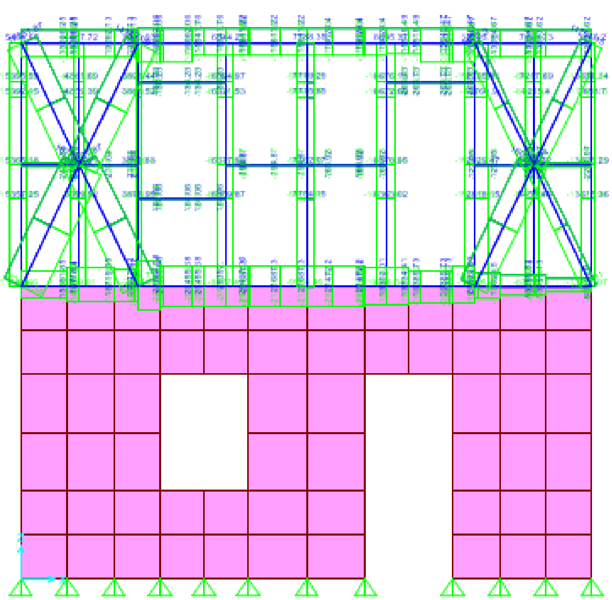
- Dead Load (DL)
- Live Load (LL)
- Earthquake Load (W)

Load Combination: For Working Stress method As per NBC 105:

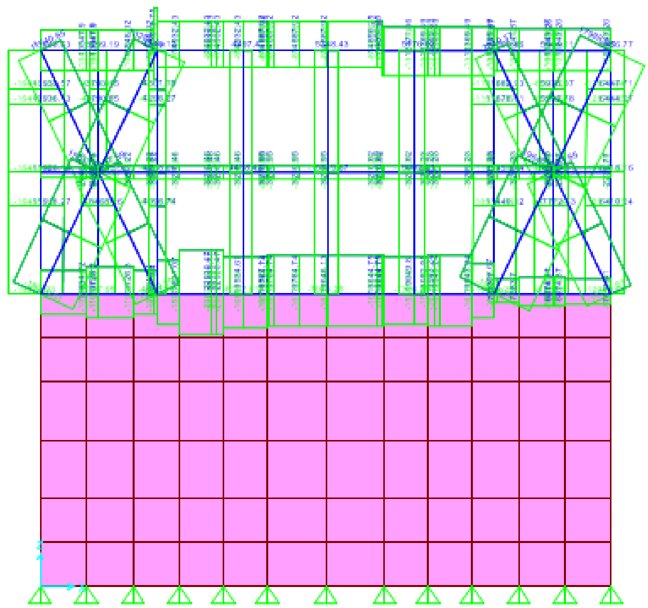
- DL+LL
- 0.7DL+W0
- 0.7DL-W0
- 0.7DL+W90
- 0.7DL-W90
- DL+LL+W0
- DL+LL-W0
- DL+LL+W90
- DL+LL-W90
- 0.7DL+W0-
- 0.7DL-W0-
- 0.7DL+W90-
- 0.7DL-W90-
- DL+LL+W0-
- DL+LL-W0-
- DL+LL+W90-
- DL+LL-W90-

Modeling output for existing building

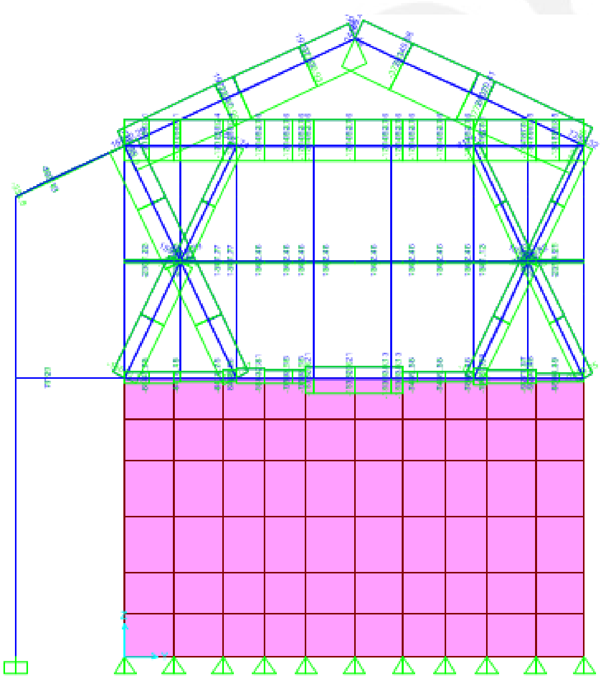
Initially, building is modeled and axial forces for different load combination is studied. The axial force develop for envelope combination is shown below.



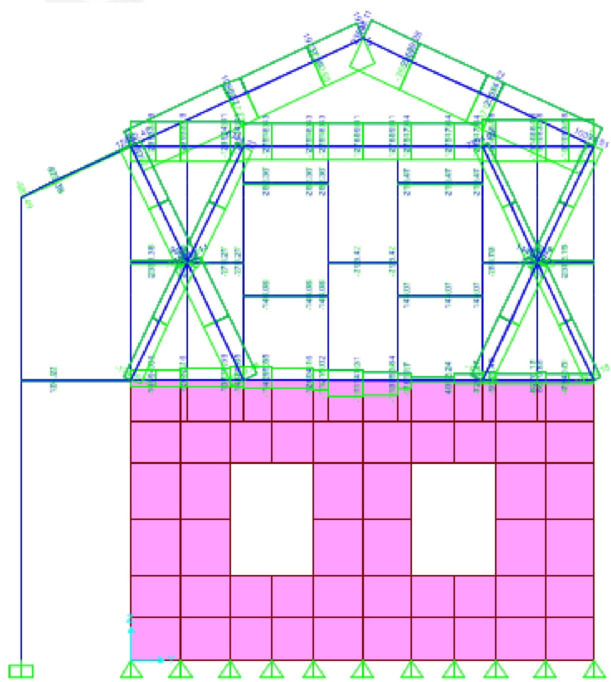
Envelope Axial Force (X-direction: Grid 1-1)



Envelope Axial Force (X-direction: Grid 2-2)



Envelope Axial Force (Y-direction: Grid A-A)



Envelope Axial Force (Y-direction-B-B)

1. Design of structure:

Timber members are mainly designed for the axial force induced due to envelope load combination of earthquake and wind load. Design force is taken from the maximum force due to earthquake and wind load. Working stress method is used for the design of timber element. Design force for different member is listed below

Member	Earthquake Load		Wind load		Design Force	
	Tension	Compression	Tension	Compression	Tension	Compression
	KN	KN	KN	KN	KN	KN
Bracing	6.94	7.14	19.64	23.42	19.64	23.42
Vertical Post	4.54	8.68	12.74	16.46	12.74	16.46
Wall Plate	11.72	13.11	31.05	21.3	31.05	21.3
Rafter	1.01	13.17	24.13	37.56	24.13	37.56
H. batten	1.51	2.06	3.21	3.51	3.21	3.51
Verandah Post	0	7.51	2.51	11.08	2.51	11.08

Design of Bracing

Induced tensile force =	19.64	KN	
Induced Compressive force =	23.42	KN	
Modulus of elasticity E =	9400	Mpa	
Allowable tensile strength (ft)=	6.9	Mpa	
Allowable Compressive Strength (fc)=	5.5	Mpa	
Allowable Shear Strength (fs)=	0.6	Mpa	
Use 50 X 115 mm Size of Bracing element			
Length of Member (L)=	1.3	m	
Width of Member (B) =	0.05	m	
Depth of Member (D) =	0.115	m	
Length / Width ratio (L/B) =	26		(<50 OK)
Sectional area of Bracing (A) =	0.0058	m2	
Check for tension			
Tensile Capacity of Member = A* ft =	39.675	KN	>19.64 OK
Check for compression			
K8 = Constant = 0.702 E /fc =	12.37		
Since L/B ration is greater than K8 , Design as long Column			
Permissible Compression Stress, fc = $0.329*E/(L/d)^2$ =	4.57	Mpa	
Compression Capacity = A* fc=	26.31	KN	>23.42 OK

Design of Vertical Post

Induced tensile force =	12.74	KN	
Induced Compressive force =	16.46	KN	
Modulus of elasticity E =	9400	Mpa	
Allowable tensile strength (ft)=	6.9	Mpa	
Allowable Compressive Strength (fc)=	5.5	Mpa	
Allowable Shear Strength (fs)=	0.6	Mpa	
Use 75 X 75 mm Size of Vertical post element			
Length of Member (L)=	2.4	m	(Leff =2.4)
Width of Member (B) =	0.075	m	
Depth of Member (D) =	0.075	m	
Length / Width ratio (L/B) =	32		(<50 OK)
Sectional area of Bracing (A) =	0.0056	m2	
Check for tension			
Tensile Capacity of Member = A* ft =	38.8125	KN	>12.74 OK
Check for compression			
K8 = Constant = 0.702 E / fc =	12.37		
Since L/B ration is greater than K8 , Design as long Column			
Permissible Compression Stress, fc = $0.329*E/(L/d)^2$ =	3.02	Mpa	
Compression Capacity = A* fc=	16.99	KN	>16.46 OK

Design of Rafter

Induced tensile force =	24.13	KN	
Induced Compressive force =	37.56	KN	
Modulus of elasticity E =	9400	Mpa	
Allowable tensile strength (ft)=	6.9	Mpa	
Allowable Compressive Strength (fc)=	5.5	Mpa	
Allowable Shear Strength (fs)=	0.6	Mpa	
Use 80 X 140 mm Size of Rafter element			
Length of Member (L)=	2.4	m	(Leff =2.4)
Width of Member (B) =	0.08	m	
Depth of Member (D) =	0.14	m	
Length / Width ratio (L/B) =	30		(<50 OK)
Sectional area of Bracing (A) =	0.0112	m2	
Check for tension			
Tensile Capacity of Member = A* ft =	77.28	KN	>24.13 OK
Check for compression			
K8 = Constant = 0.702 E / fc =	12.37		
Since L/B ration is greater than K8 , Design as long Column			
Permissible Compression Stress fc = $0.329*E/(L/d)^2$ =	3.44	Mpa	
Compression Capacity = A* fc=	38.49	KN	>37.56 OK

Design of Wall Plate:

Induced tensile force =	31.05	KN	
Induced Compressive force =	21.3	KN	
Modulus of elasticity E =	9400	Mpa	
Allowable tensile strength (ft)=	6.9	Mpa	
Allowable Compressive Strength (fc)=	5.5	Mpa	
Allowable Shear Strength (fs)=	0.6	Mpa	
Use 75 X 100 mm Size of Wall plate element			
Length of Member (L)=	0.5	m	(Leff =2.4)
Width of Member (B) =	0.075	m	
Depth of Member (D) =	0.1	m	
Length / Width ratio (L/B) =	6.66667		(<50 OK)
Sectional area of Bracing (A) =	0.0075	m2	
Check for tension			
Tensile Capacity of Member = A* ft =	51.75	KN	>31.05 OK
Check for compression			
K8 = Constant = 0.702 E / fc =	12.37		
Since L/B ratio is Smaller than K8 and 11 , Design as Short Column			
Permissible Compression Stress fc = $0.329*E/(L/d)^2$ =	5.50	Mpa	
Compression Capacity = A* fc=	41.25	KN	>21.3 OK

Design of Horizontal Batten:

Induced tensile force =	3.21	KN	
Induced Compressive force =	3.51	KN	
Modulus of elasticity E =	9400	Mpa	
Allowable tensile strength (ft)=	6.9	Mpa	
Allowable Compressive Strength (fc)=	5.5	Mpa	
Allowable Shear Strength (fs)=	0.6	Mpa	
Use 25 X 75 mm Size of Batten element			
Length of Member (L)=	0.975	m	(Leff =2.4)
Width of Member (B) =	0.025	m	
Depth of Member (D) =	0.075	m	
Length / Width ratio (L/B) =	39		(<50 OK)
Sectional area of Bracing (A) =	0.0019	m2	
Check for tension			
Tensile Capacity of Member = A* ft =	12.9375	KN	>3.21 OK
Check for compression			
K8 = Constant = 0.702 E / fc =	12.37		
Since L/B ration is greater than K8 , Design as long Column			
Permissible Compression Stress = $0.329*E/(L/d)^2$ =	2.03	Mpa	
Compression Capacity = A* fc=	3.81	KN	>3.51 OK

Design of Verandah Post:

Induced tensile force =	2.51	KN	
Induced Compressive force =	11.08	KN	
Modulus of elasticity E =	9400	Mpa	
Allowable tensile strength (ft)=	6.9	Mpa	
Allowable Compressive Strength (fc)=	5.5	Mpa	
Allowable Shear Strength (fs)=	0.6	Mpa	
Use 100 X 100 mm Size of Verandah post element			
Length of Member (L)=	3	m	(Leff =2.4)
Width of Member (B) =	0.1	m	
Depth of Member (D) =	0.1	m	
Length / Width ratio (L/B) =	30		(<50 OK)
Sectional area of Bracing (A) =	0.0100	m2	
Check for tension			
Tensile Capacity of Member = A* ft =	69	KN	>2.51 OK
Check for compression			
K8 = Constant = 0.702 E / fc =	12.37		
Since L/B ration is greater than K8 , Design as long Column			
Permissible Compression Stress = $0.329*E/(L/d)^2$ =	3.44	Mpa	
Compression Capacity = A* fc=	34.36	KN	>11.08 OK

Design of Connection

Connection of Post and Wall Plate

Tensile Force on the post =	12.74	KN	
Compression Force on the post =	14.46	KN	
Yield Strength of Steel plate (fy)=	250	Mpa	
Strength of Plate in tearing (ft)=	150	Mpa	
Shear Strength of bolt (tb) =	100	Mpa	
Use 12 mm dia bolt			
Diameter of the Bolt (d) =	12	mm	
Shear Area of Bolt in Single Shear =	113.1	mm2	
Capacity of one bolt in Single Shear =	11.31	KN	
	1.1264		
Number of Bolt required =	6		
Use 2-12 mm Bolt			
Check for tearing of Plate			
Thickness of plate =	4	mm	
Effective length (l-n*d0) =	126	mm	
Tearing Capacity =	75.6	KN	>12.74O K

Connection of Bracing, Post and Wall plate

Tensile force of the post =	12.74	KN	
Compression Force on the post =	16.76	KN	
Tensile force on the Bracing =	19.64	KN	
Angle of Bracing =	64	deg	
Compression force on the Bracing =	23.42	KN	
Net vertical Force in Connection =	37.81	KN	
Net Horizontal Force in Connection =	8.61	KN	
Yield Strength of Steel plate (fy)=	250	Mpa	
Strength of Plate in tearing (ft)=	150	Mpa	
Shear Strength of bolt (tb) =	100	Mpa	
Use 12 mm dia bolt			
Diameter of the Bolt (d) =	12	mm	
Shear Area of Bolt in Single Shear =	113.1	mm2	
Capacity of one bolt in Single Shear =	11.31	KN	
	1.7365		
Number of Bolt required in Bracing =	6	Nos	
	1.1264		
Number of Bolt required in Post =	6	Nos	
	3.3431		
Number of Bolt in wall Plate =	2	nos	
Use 2-12 mm dia @ bracing and post and 4-12 mm dia bolt in Wall plate			
Check for tearing of Plate			
Thickness of plate =	4	mm	
Effective length (l-n*d0) =	76	mm	
Tearing Capacity =	45.6	KN	>37.87 OK

Connection of Rafter and Wall plate

Tensile Force on the Rafter =	24.13	KN	
Compression Force on the Rafter =	37.56	KN	
Yield Strength of Steel plate (fy)=	250	Mpa	
Strength of Plate in tearing (ft)=	150	Mpa	
Shear Strength of bolt (tb) =	100	Mpa	
Use 12 mm dia bolt			
Diameter of the Bolt (d) =	12	mm	
Shear Area of Bolt in double Shear =	226.2	mm2	
Capacity of one bolt in double Shear =	22.62	KN	
Number of Bolt required =	1.06678		
Use 1-12 mm Bolt			
Check for tearing of Plate			
Thickness of plate =	4	mm	
Effective length (l-n*d0) =	48	mm	
Tearing Capacity =	28.8	KN	>24.13OK

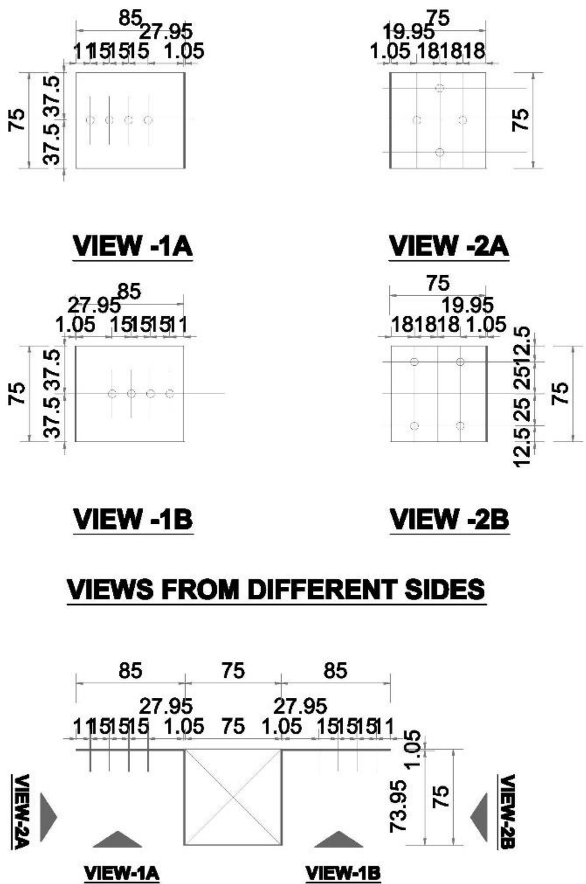
Connection of Horizontal Batten and post

Tensile Force on the post =	3.21	KN	
Compression Force on the post =	3.51	KN	
Yield Strength of Steel plate (fy)=	250	Mpa	
Strength of Plate in tearing (ft)=	150	Mpa	
Shear Strength of bolt (tb) =	100	Mpa	
Use 12 mm dia bolt			
Diameter of the Bolt (d) =	12	mm	
Shear Area of Bolt in Single Shear =	113.1	mm2	
Capacity of one bolt in Single Shear =	11.31	KN	
	0.2838		
Number of Bolt required =	3		
Use 1-12 mm Bolt			
Check for tearing of Plate			
Thickness of plate =	4	mm	
Effective length (l-n*d0) =	63	mm	
Tearing Capacity =	37.8	KN	>3.21OK

Connection of Wall plate and Masonry wall

Maximum Uplift force on Wind =	141.906	KN
At least 4 vertical post will be there and 8 nail at each vertical		
Number of Nails =	32	
Diameter of Nail =	3.55	mm
Shear Strength =	100	Mpa
Shear Area of each nail =	9.90	mm2
Shear Capacity of each nail =	0.99	KN
Total Capacity =	31.7	KN
Residual Uplift Force =	110.23	KN
Use Gabion wire , of 3.25 mm dia		
Diameter of gabion =	3.25	mm
Section Area of gabion wire =	16.59	mm2
Tensile Strength of the Gabion =	140	Mpa
Capacity at Each level =	2.323	KN
Number of Gabion Required =	47.46	Nos
Total Length of Wall =	21.6	m
Spacing required for Gabion =	455.2	mm

Provide gabion of 3.25 mm (10 Gauge) at the spacing of 450 mm C/C throughout the wall

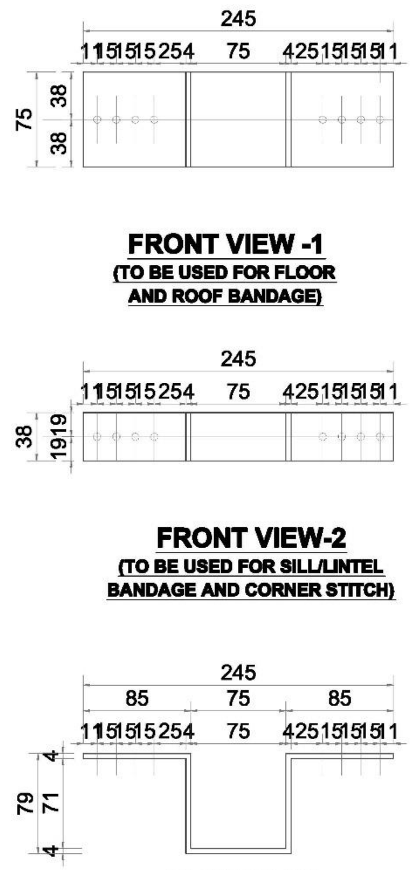


PLAN VIEW

CGI SHEET DETAIL

(ALTERNATE OPTION FOR METAL STRIP)

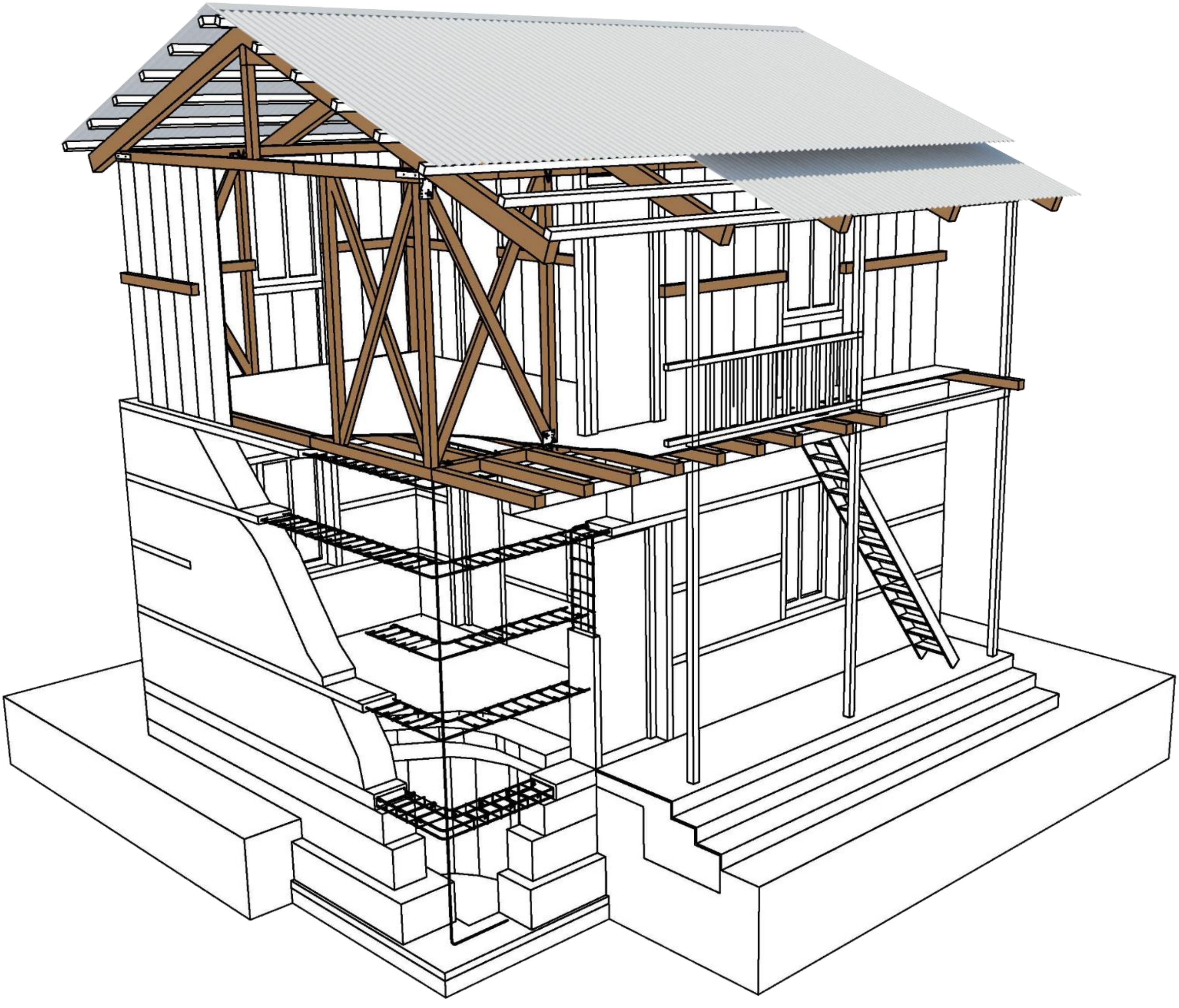
SCALE = 1:5



PLAN VIEW

METAL STRIP DETAIL

SCALE = 1:5



नेपाल सरकार
राष्ट्रिय पुनर्निर्माण प्राधिकरण
आवास पुनर्निर्माण कार्यक्रम
सिंहदरवार, काठमाडौं
फोन नं.: ०१४२००२६६, ४२१११०३
इमेलः info@nra.gov.np